Comprehensive Statistical Model Evaluation

Traditional, Bayesian, & Information-Theoretic Methods + User-Friendly Software



DENVER 2025 ANNUAL MEETING ncme

Introductions



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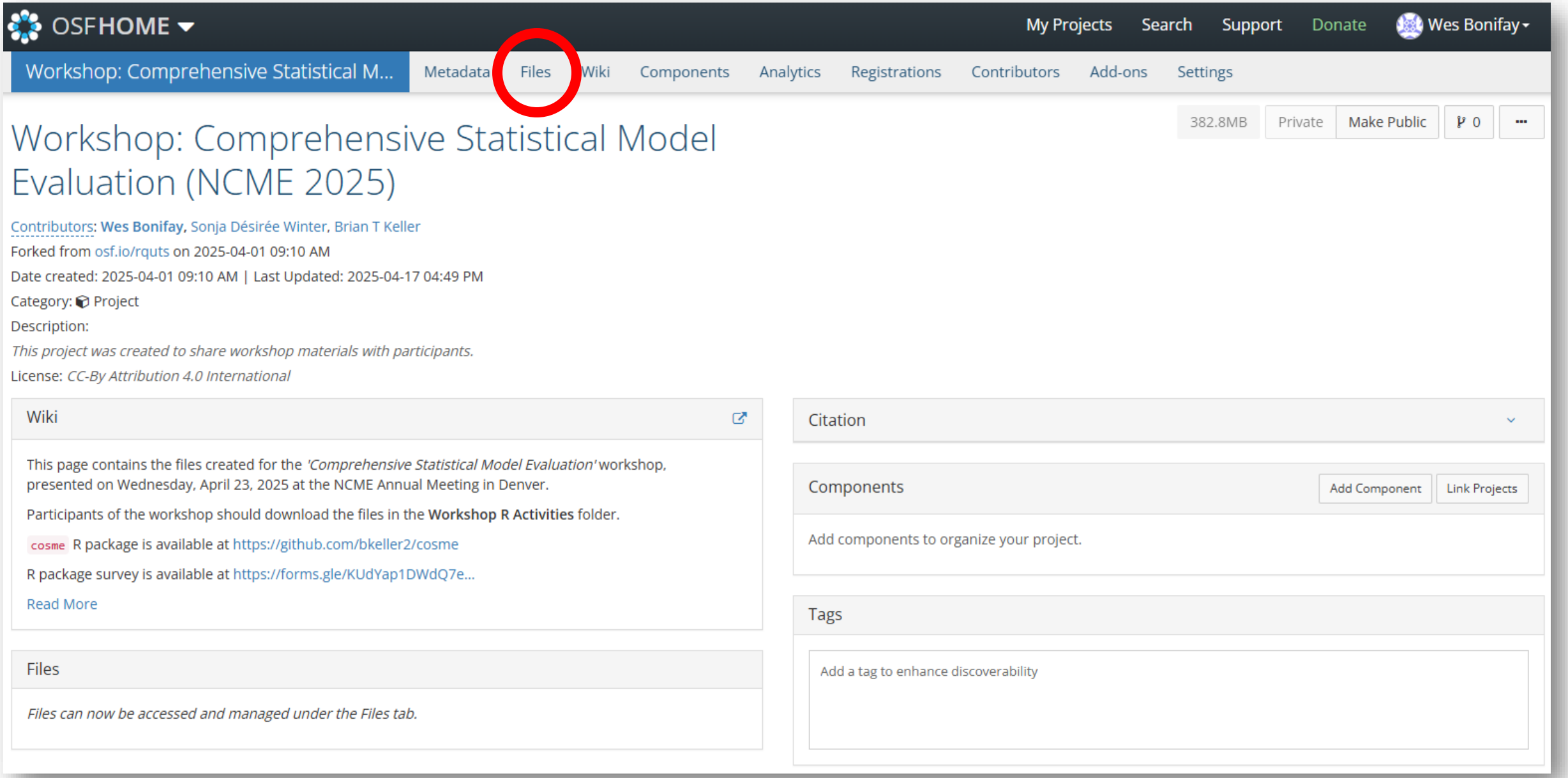
Statistics, Measurement, & Evaluation in Education
University of Missouri

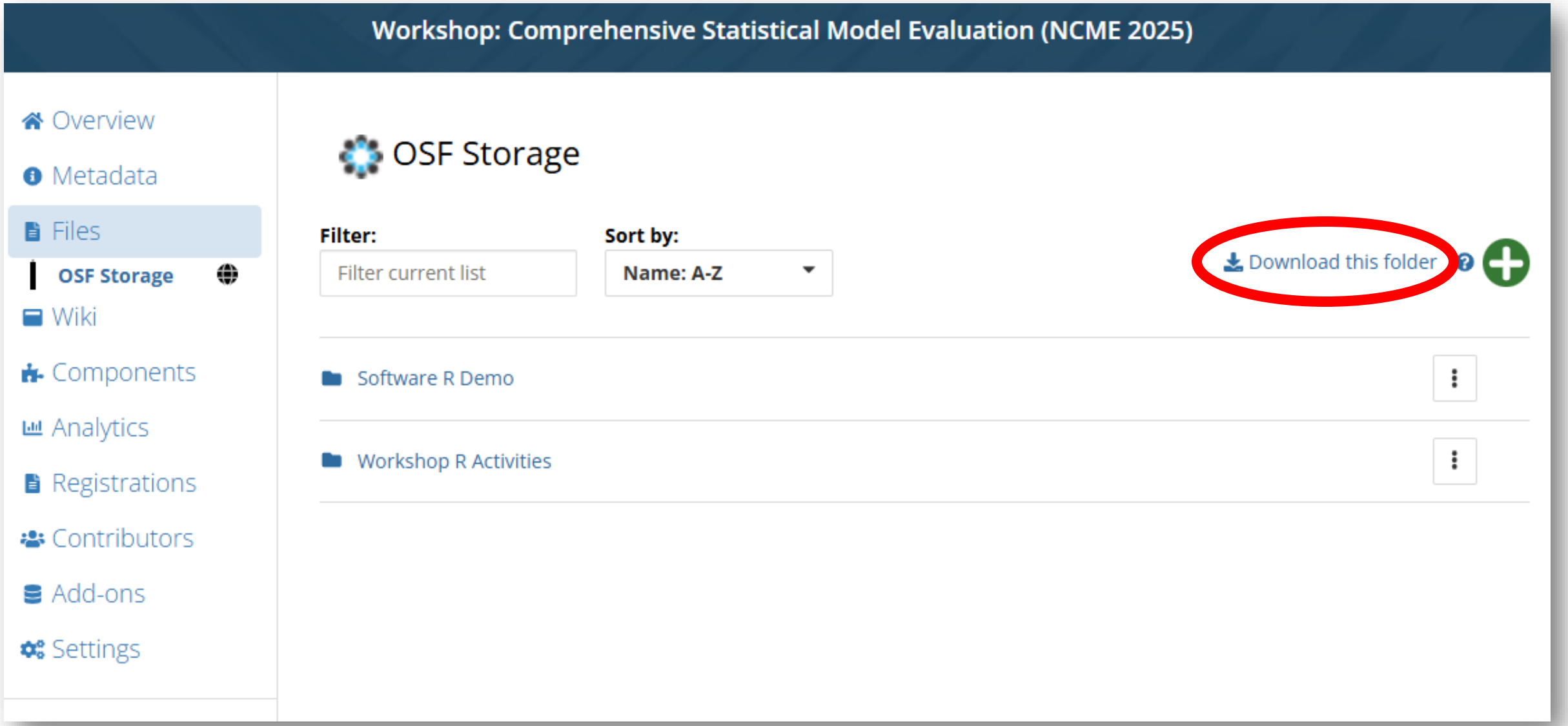
Comprehensive Statistical Model Evaluation

Resources for this workshop

Activities, data, and additional materials are available on OSF:

https://osf.io/ct4vr/


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This page contains the files created for the 'Comprehensive Statistical Model Evaluation' workshop, presented on Wednesday, April 23, 2025 at the NCME Annual Meeting in Denver.
Participants of the workshop should download the files in the Workshop R Activities folder.
cosme R package is available at https://github.com/bkeller2/cosme
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What is a scientific model?

"A model is a representation of reality intended to explain the behavior of some aspect of it."

~ Miller & Starr (1969)

"This simplified version of reality is called a model. It is a most convenient word, for it is [our] only recourse. The human mind is incapable of thinking other than about models."

~ Isaacs (1975)

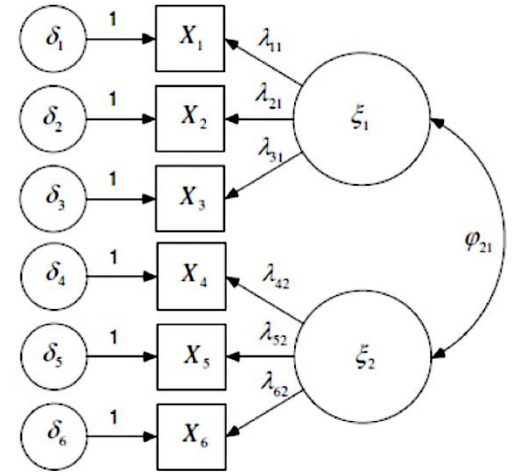


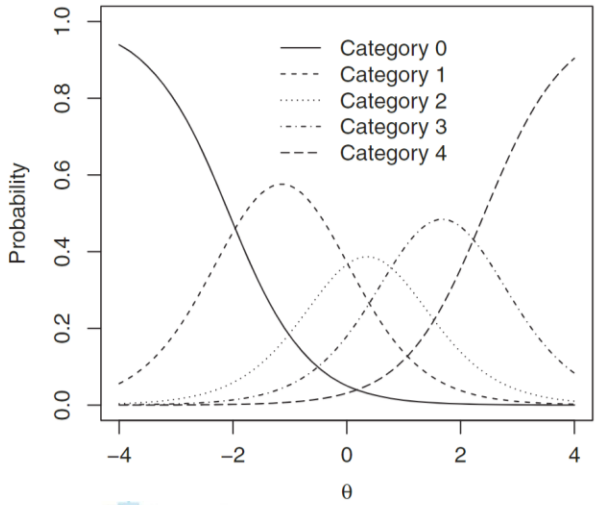
Examples: regression analysis, structural equation modeling, item response theory, network modeling, etc.

What is a *statistical* model?

"Statistical models are succinct representations of relations between variables, which are usually interpretable as reflecting theories and hypotheses about a phenomenon of interest (technically, the data-generating process)."

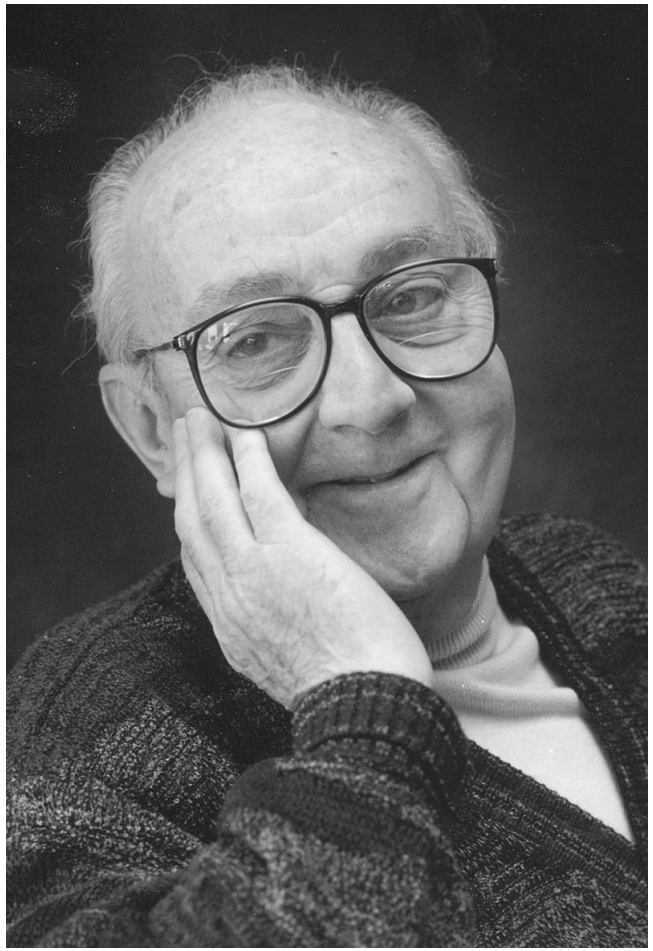
~ Fanelli, 2024





Examples: regression analysis, structural equation modeling, item response theory, diagnostic classification modeling, latent class/profile analysis, network modeling, etc.

- Basically, everything here



All models are wrong, but some are useful

~ George E. P. Box (1978)

What makes a model useful?

Example: Goodness-of-fit testing

Frequentist Statistics

Example: Fitting propensity analysis

Information Theory

Bayesian Inference

Example: Predictive model checking

A frequentist on Bayesians: “If they would only do as [Bayes] did and publish posthumously, we should all be saved a lot of trouble” ~ Kendall

A Bayesian on frequentists: “There is a valid defense of using non-Bayesian methods, namely incompetence.” ~ Skilling

An information-theorist on Bayesians ***and*** frequentists: “[Their] deductions may be entertaining but are quite irrelevant to the task at hand.” ~ Rissanen

Outline

1. Traditional Model Evaluation
2. Bayesian Model Evaluation
3. Information-theoretic Model Evaluation
4. Summary
5. Software

1. Traditional Model Evaluation



Example: Goodness-of-fit testing

Frequentist Statistics

Example: Fitting propensity analysis

Information Theory

Bayesian Inference

Example: Predictive model checking

Goodness-of-fit testing

For maximum likelihood-based estimators, we can formally evaluate the *goodness-of-fit* of a model to the observed data

- How accurately does the model-implied covariance matrix reflect the observed covariance matrix?

- $H_0 : \mathbf{\Sigma} = \mathbf{R}$
- H_A : $\mathbf{\Sigma}$ is any symmetric, positive definite matrix

Goodness-of-fit testing

Under H_0, we can compute a chi-square test statistic: $\chi^2 = (N-1)\hat{F}$

with $df = \frac{n(n+1)}{2} - n - np + \frac{p(p-1)}{2}$

where

- N = sample size
- $\hat{F}$ = the value of the discrepancy function at its minimum

- If χ^2 is significant (typically meaning the *p*-value < .05), then we must reject H_0, i.e., reject the proposed model with p factors
- If χ^2 is not significant (typically meaning the *p*-value > .05), then we cannot reject H_0, i.e., we accept the model with p factors
- If the model is a perfect representation of the data, then $\chi^2 = df$

Goodness-of-fit Testing

- Is it reasonable to test the *exact* fit of a statistical model when we know it's probably wrong?
  - What we need to know is if our model is *importantly* wrong (Box, 1976)
- Main types of approximate fit indices
  - *Absolute* fit indices: How well does the model explain the data without a reference model?
  - *Incremental/relative/comparative* indices: How well does the model explain the data compared to a baseline (independence) model?
  - *Noncentrality* indices: estimate the degree to which the exact fit hypothesis is false (i.e., a sort of effect size)

Some Approximate Fit Tests

- Root Mean Square Error of Approximation (RMSEA; Steiger, 1990) and its 90% confidence interval
  - Absolute, noncentrality fit index
  - Badness-of-fit index, so 0 is the best result
- Comparative Fit Index (CFI; Bentler, 1990)
  - Incremental, noncentrality fit index
  - Goodness-of-fit index ranging from 0 to 1, where 1 is the best result
- Standardized Root Mean Square Residual (SRMR; Jöreskog & Sörbom, 1981)
  - Absolute fit index
  - Badness-of-fit index, so 0 is best result







RMSEA

$$RMSEA = \sqrt{\frac{\max(0, \chi^2 - df_M)}{df_M(N-1)}}$$

- $\max(0, \chi^2 - df_M)$ means that the noncentrality parameter estimate is forced to be at least 0. This will happen when $\chi^2 \leq df_M$
  - Thus, RMSEA can be zero (best case), while the χ^2 is not zero (but is smaller than the degrees of freedom)
- We can form a one-sided 90% confidence interval (using the 0th and 90th percentile) around the RMSEA
- Different cutoffs for separating good from poor fit have been proposed: ≤ .05, .06, .08, .10… None work in all situations
  - RMSEA is inflated (indicating poor fit too often) for models with very few degrees of freedom, so not recommended in those cases

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CFI

- What is the baseline model?
  - Basically, the opposite of the unrestricted model!
  - Chi-square compares hypothesized model to “best” model
  - CFI compares hypothesized model to “worst” model
- Interpretation:
  - CFI = .90 means that the hypothesized model reduced the size of the noncentrality parameter by 90% compared to the baseline model
- Typical cutoff is ≥ .95 (Hu & Bentler, 1999)
  - Can also use Tucker-Lewis Index (TLI), which is less sensitive to sample size
  - Both CFI and TLI are inflated when variables are highly correlated



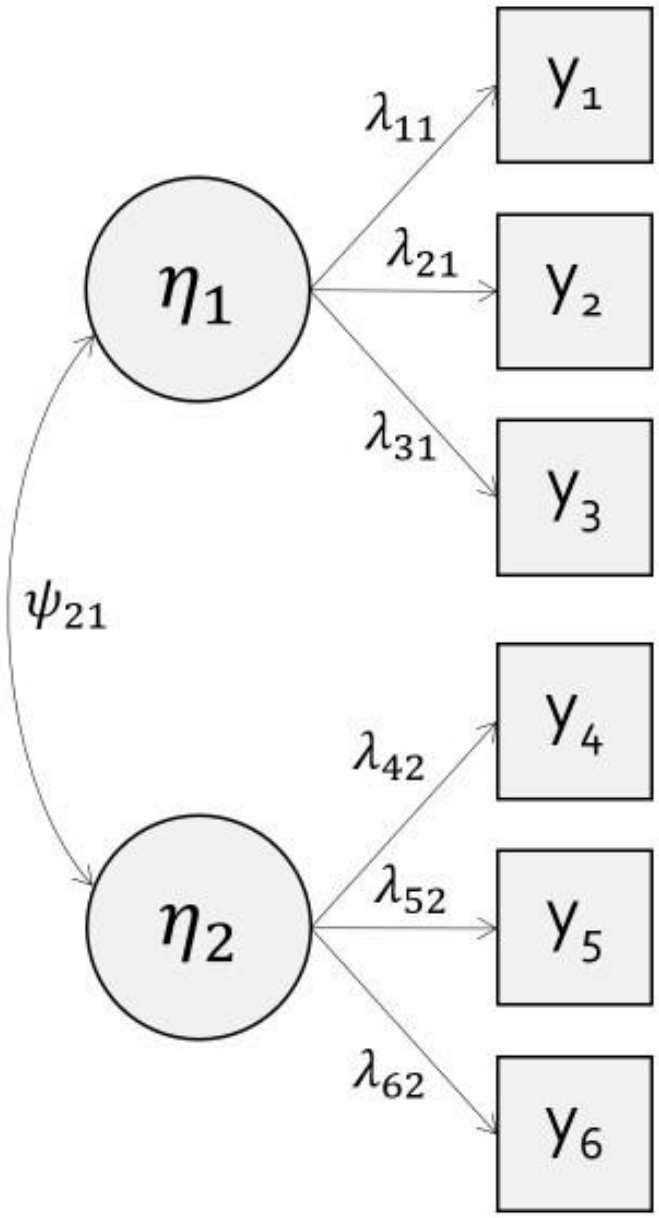
Activity Time!

Let’s examine CFA fit statistics

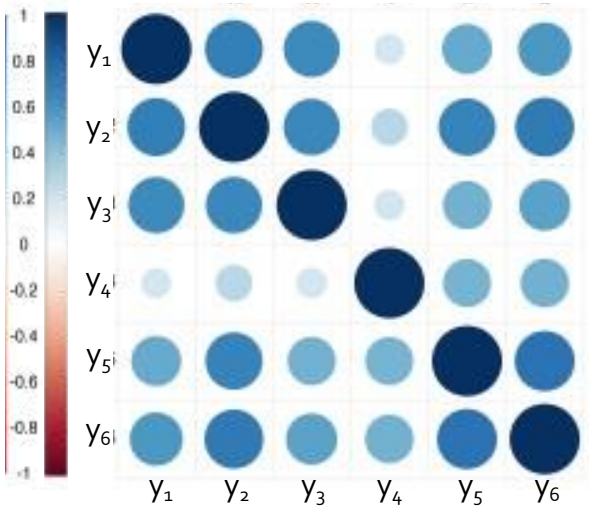
Load the activity1.R file



A. Statistical Model

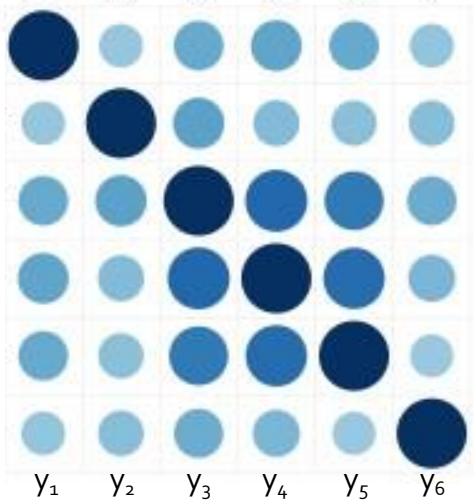


B. Original Data



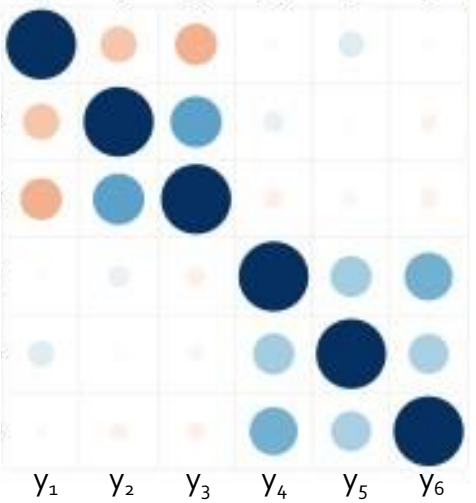
	η_1	η_2
λ_{11}	.77	
λ_{21}	.90	
λ_{31}	.73	
λ_{42}		.50
λ_{52}		.82
λ_{62}		.90
ψ_{21}	.84	
CFI	.96	

C. Fit Replication #1



	η_1	η_2
λ_{11}	.60	
λ_{21}	.57	
λ_{31}	.90	
λ_{42}		.91
λ_{52}		.83
λ_{62}		.52
ψ_{21}	.96	
CFI	.96	

D. Fit Replication #2



	η_1	η_2
λ_{11}	.42	
λ_{21}	-.63	
λ_{31}	-.86	
λ_{42}		.70
λ_{52}		.49
λ_{62}		.68
ψ_{21}	.09	
CFI	.95	

What happened?

Psychological Review
2000, Vol. 107, No. 2, 358–367

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0033-295X/00/$5.00 DOI: 10.1037//0033-295X.107.2.358

THEORETICAL NOTES

How Persuasive Is a Good Fit? A Comment on Theory Testing

Seth Roberts
University of California, Berkeley

Harold Pashler
University of California, San Diego

Quantitative theories with free parameters often gain credence when they closely fit data. This is a mistake. A good fit reveals nothing about the flexibility of the theory (how much it cannot fit), the variability of the data (how firmly the data rule out what the theory cannot fit), or the likelihood of other outcomes (perhaps the theory could have fit any plausible result), and a reader needs all 3 pieces of information to decide how much the fit should increase belief in the theory. The use of good fits as evidence is not supported by philosophers of science nor by the history of psychology; there seem to be no examples of a theory supported mainly by good fits that has led to demonstrable progress. A better way to test a theory with free parameters is to determine how the theory constrains possible outcomes (i.e., what it predicts), assess how firmly actual outcomes agree with those constraints, and determine if plausible alternative outcomes would have been inconsistent with the theory, allowing for the variability of the data.

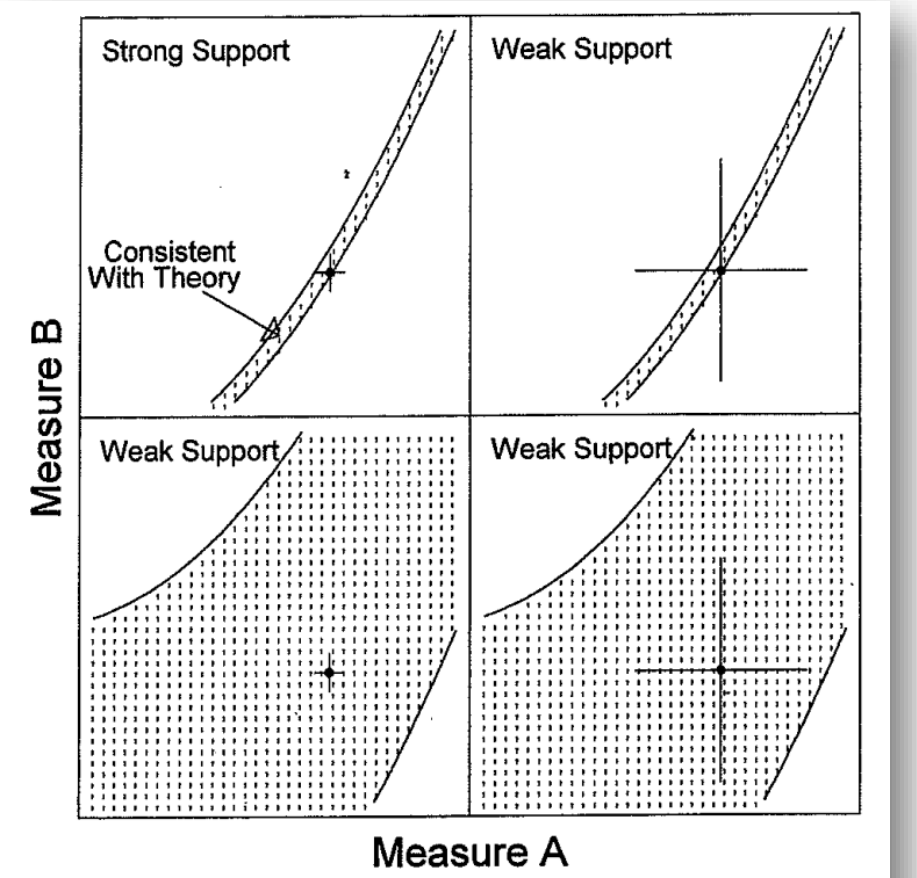


Figure 1. Four possible relationships between theory and data. Measures A and B are measures of behavior. For both measures, the axes cover the whole range of plausible values. The dotted areas indicate the range of outcomes that would be consistent with the theory. The error bars indicate standard errors. In every case, the theory can closely fit the data, but only when both theory and data provide substantial constraints does this provide significant evidence for the theory.

Limitations of Goodness-of-Fit Testing

Good fit:

1. Cannot tell us what a theory/model *predicts*
2. Does not reflect how well the *data* agree with those predictions
3. Ignores the a priori likelihood that the theory/model will fit *any plausible result*

Roberts & Pashler (2000)

Clearly, then, showing that a theory fits data is not enough. By itself, it is nearly meaningless. Because of the flexibility of many theories, the variability of measurements, and the simplicity of most psychological data functions, it is often quite possible that the theory could fit any plausible outcome to within the precision of the data. The reader has no way of knowing which panel of Figure 1 the evidence resembles.

Thus we did not find any support within the philosophy of science for the use of good fits to support theories.

So we did not find any support in the history of psychology for the use of good fits to support theories.

Roberts & Pashler (2002)

When used to evaluate complex psychological theories, goodness-of-fit tests have been too easy to pass. The theories are too flexible, the data too variable, and contradictory outcomes too implausible. We fear we did not state this bluntly enough.

Even good tools can be misused. In Roberts and Pashler (2000), we were not saying that the use of good fits as evidence for complex theories had been misused. We were saying—perhaps too politely, perhaps not—that the practice is rotten to the core, in the sense that a taxi-driver test that can be passed by writing your name is rotten to the core.

Any questions about traditional (frequentist) model evaluation?

2. Bayesian Model Evaluation



Example: Goodness-of-fit testing

Frequentist Statistics

Example: Fitting propensity analysis

Information Theory

Bayesian Inference

Example: Predictive model checking

Why is the sidewalk wet?



Prior Possibilities







New Evidence





Why is the sidewalk wet?



Posterior Possibilities







New Evidence





Bayes Theorem (1763)

New evidence: Likelihood of data values given parameter values

Prior probability of parameter values

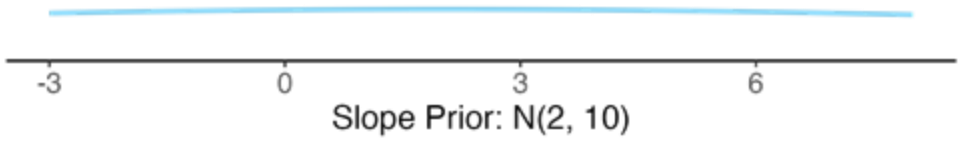
Posterior probability of parameter values, given the data

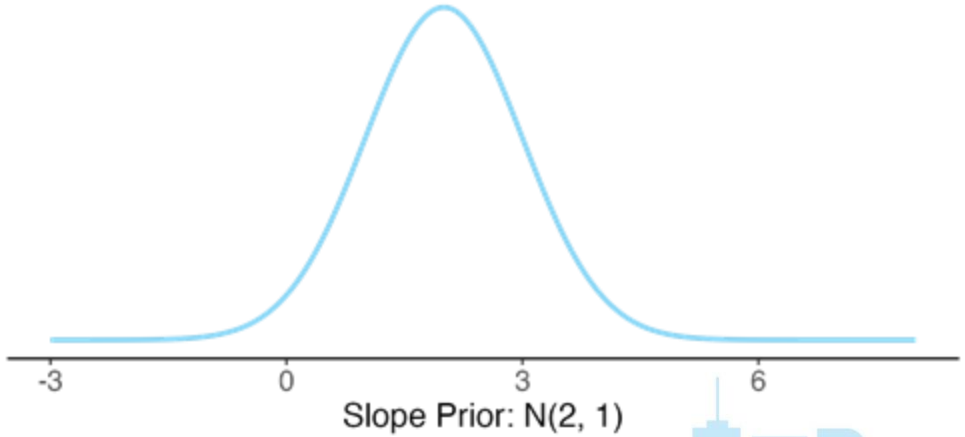
$$p(\theta|D) = \frac{p(D|\theta) \times p(\theta)}{p(D)}$$

Marginal probability of the data values

How to use prior probabilities in statistical models

- In statistical modeling, focus is on estimating parameters
- Prior probabilities about the values that those parameters might take on can be expressed through distributions
  - Unsure about probable parameter values? Specify a distribution that covers a vast range of values (diffuse prior)
  - Have some idea about probable parameter values? Specify a narrow distribution (informative prior)





How to use prior probabilities in statistical models

Example with simple linear regression

- We want to figure out if we can use a person's height (in inches) to predict their weight (in lbs)

Slope: How much does predicted weight increase if height goes up by 1 inch

Random variation around predicted weight

Observed weight

$$y_{weight} = \beta_0 + \beta_1 X_{height} + \epsilon$$

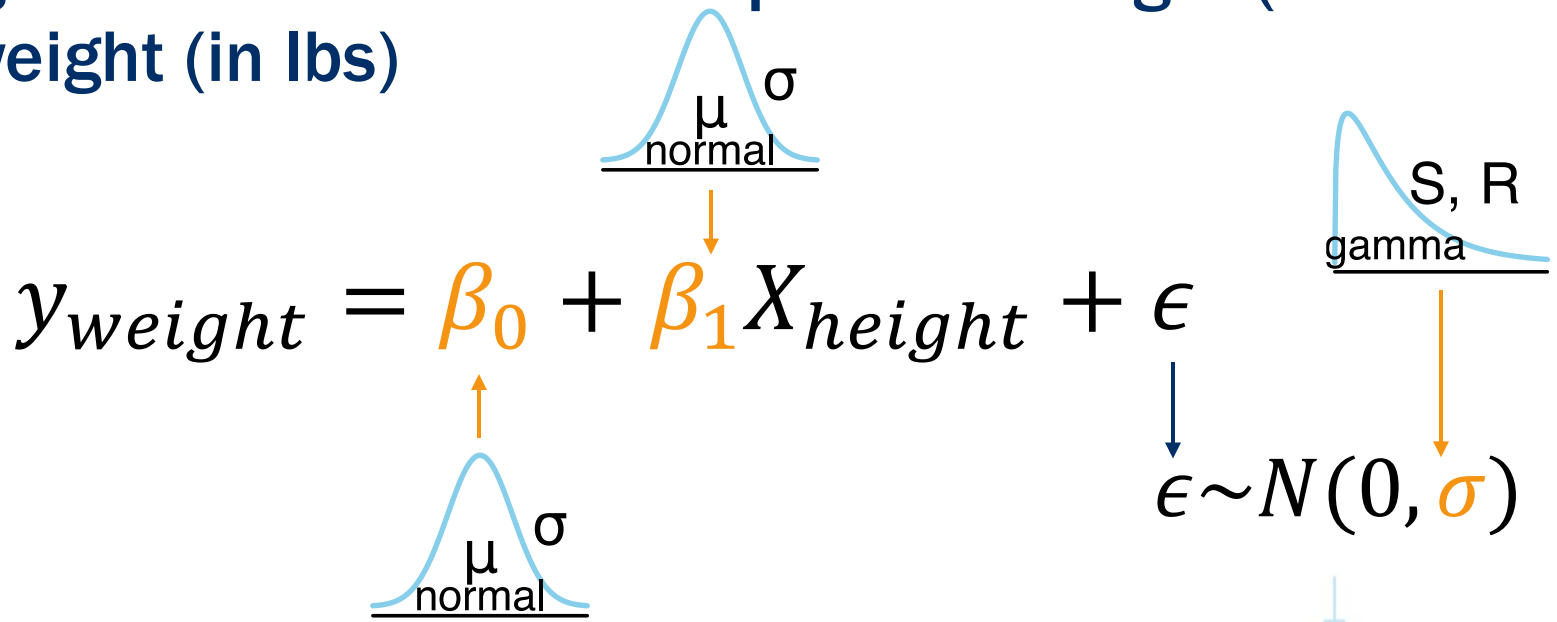
Baseline weight of person who is 0 inches tall

$$\epsilon \sim N(0, \sigma)$$

How to use prior probabilities in statistical models

Example with simple linear regression

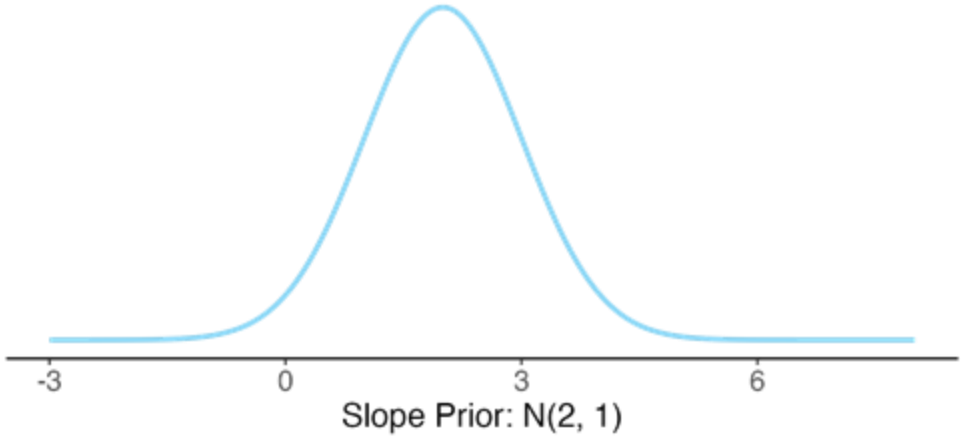
- We want to figure out if we can use a person's height (in inches) to predict their weight (in lbs)

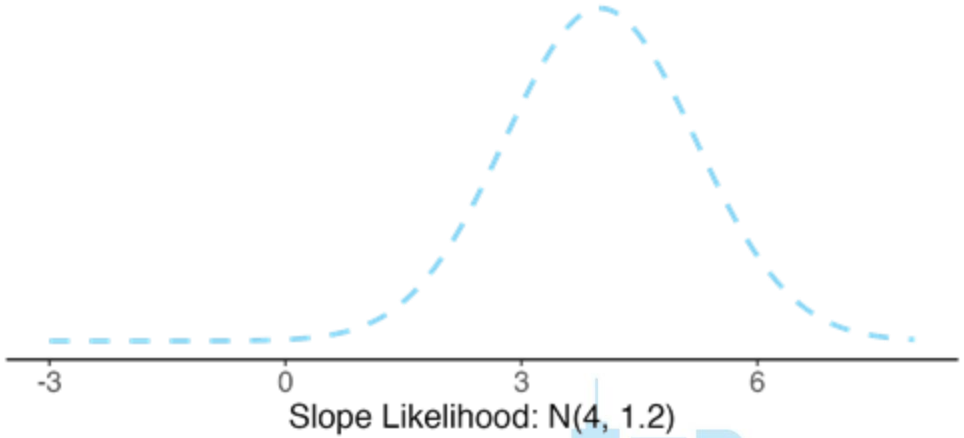


Combining prior probabilities with data

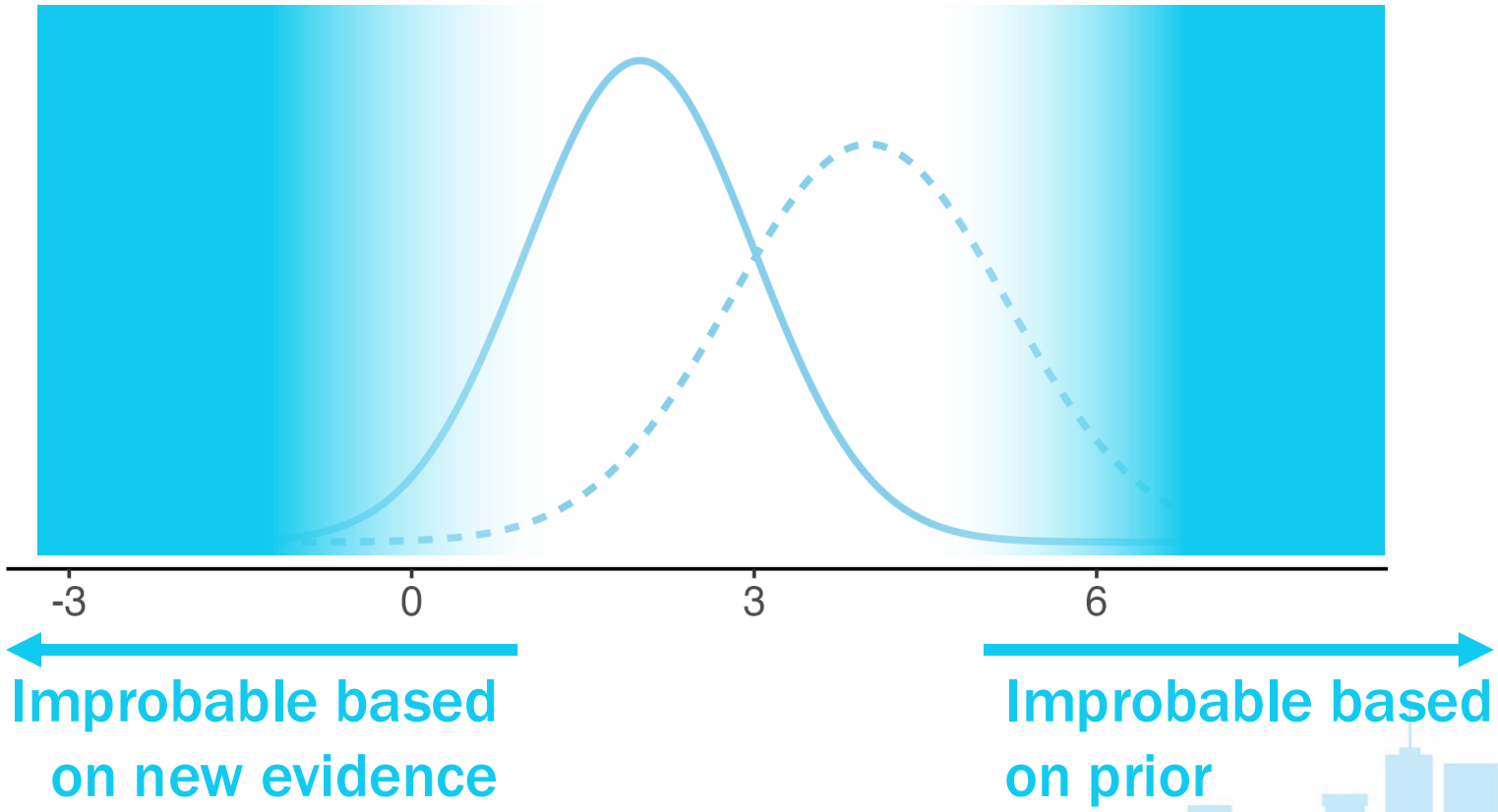
- **Prior probability:** for every inch of additional height, I expect weight to increase by 2 lbs., with a standard deviation of 1 to express uncertainty.

- **New evidence:** Using height and weight data from 30 randomly selected people, the new evidence suggests that weight is expected to increase by about 4 lbs. for every inch of additional weight.

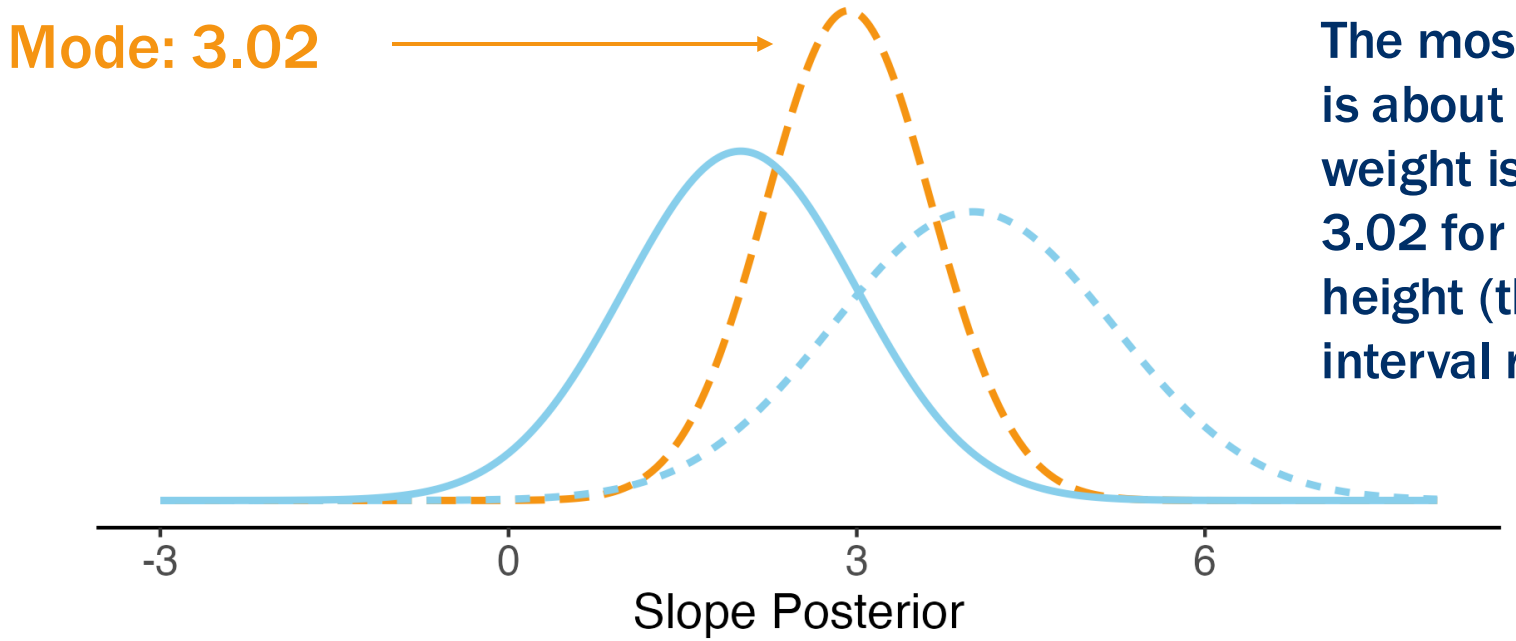




Finding the balance between prior probabilities and new evidence



Updating the posterior probabilities



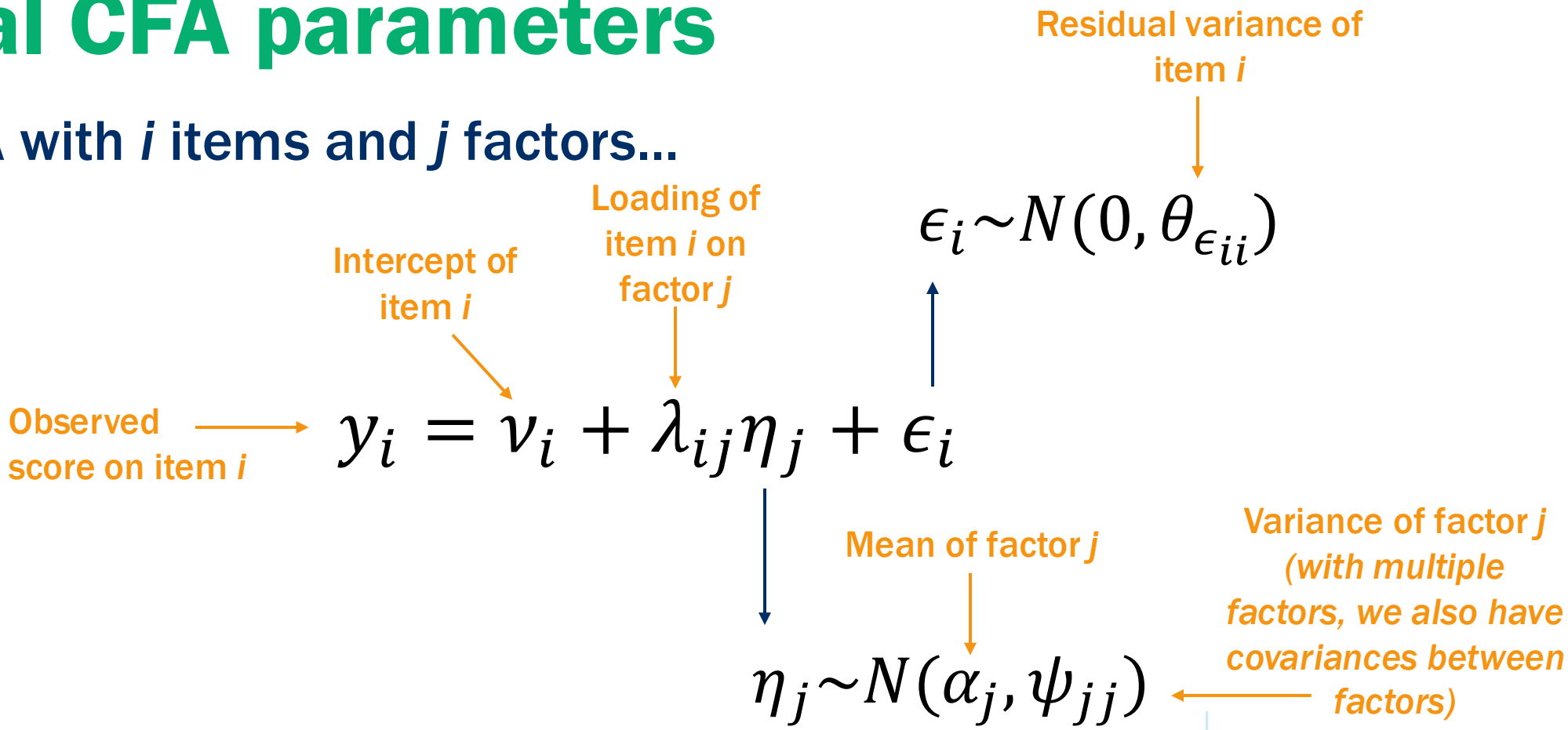
The most credible value of the slope is about 3.02, which means that weight is expected to increase by 3.02 for every inch of additional height (the 95% most credible value interval ranges from 1.60 to 4.39).

Steps of Bayesian Data Analysis

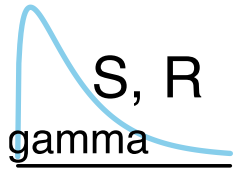
1. Identify data relevant to the research question
   - Responses to survey items
2. Describe a statistical model for the data
   - Confirmatory Factor Analysis
3. Specify prior distributions for all parameters in the model, which must seem reasonable to the audience of the analysis
   - Priors on factor loadings, item intercepts, residual variances, factor means, factor variances, factor covariances
4. Use Bayesian estimation to re-allocate credibility across parameter values (i.e., form posterior distributions)
5. Check that posterior predictions mimic the data with reasonable accuracy (i.e., conduct a “posterior predictive check”)

Typical CFA parameters

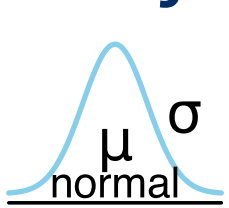
For a CFA with *i* items and *j* factors…

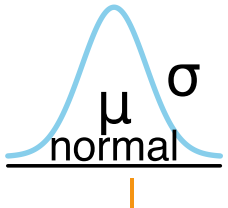


Typical priors for CFA parameters



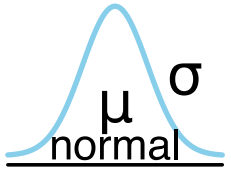
For a CFA with *i* items and *j* factors...

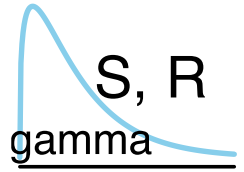




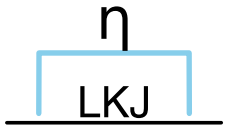
$$\epsilon_i \sim N(0, \theta_{\epsilon_{ii}})$$

$$y_i = \nu_i + \lambda_{ij}\eta_j + \epsilon_i$$



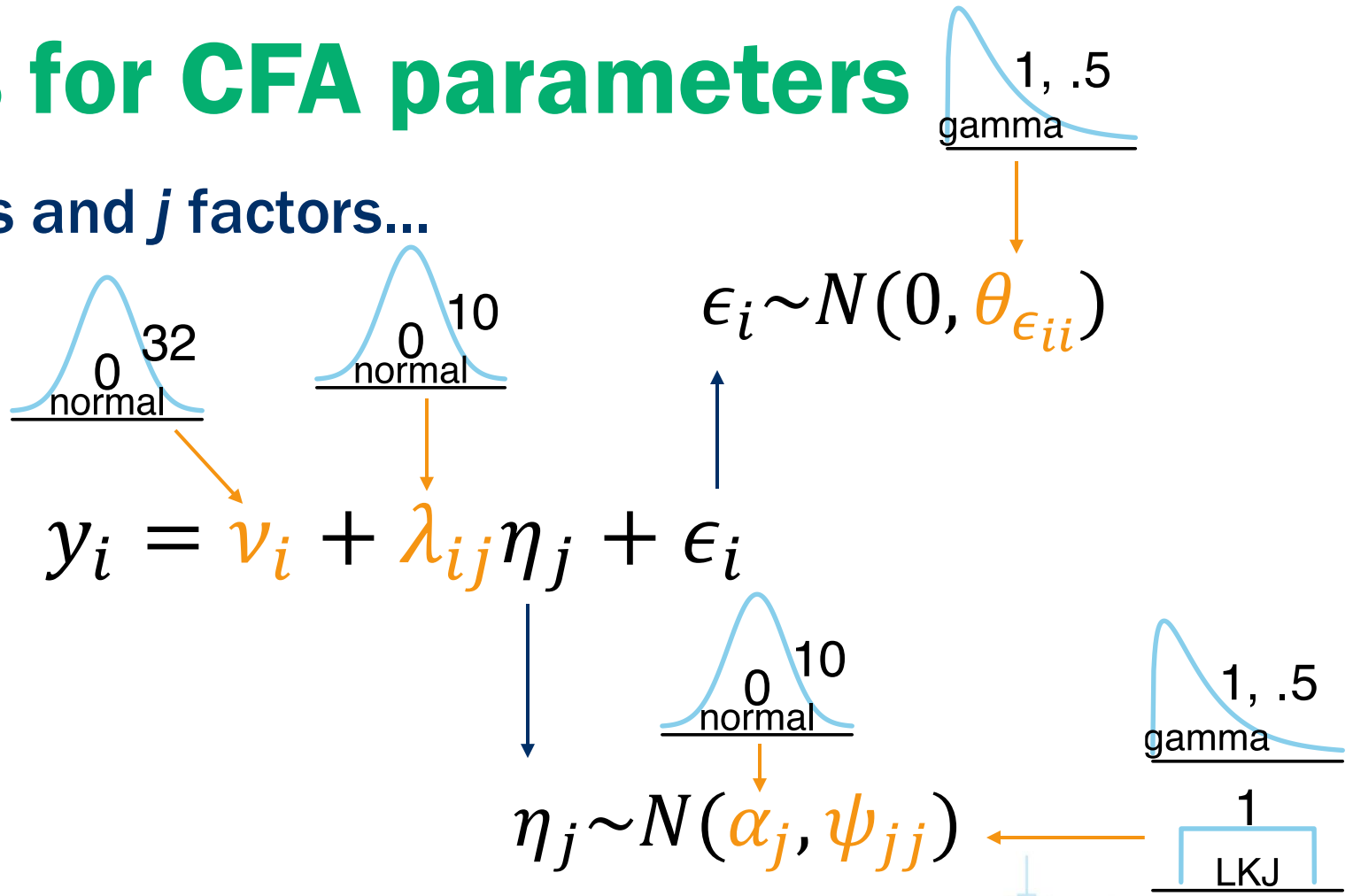


$$\eta_j \sim N(\alpha_j, \psi_{jj})$$



Typical priors for CFA parameters

For a CFA with *i* items and *j* factors...



Bayesian Estimation of CFAs

- There are many software options available for estimating CFAs using Bayesian methods
  - Two most common options are MPlus and the R package blavaan
- The posterior distribution is (typically) approximated by drawing many samples
  - This only requires that the prior probability of any parameter value can be computed and that the likelihood can be computed for any combination of parameter and data values
- Sampling from the posterior distribution can be done with different sampling algorithms
  - MPlus uses a variety of the Gibbs sampler
  - blavaan uses the no-u-turn sampler (NUTS)

Interpreting Posterior Distributions

- Before interpreting the posterior distributions, convergence of the sampling algorithm needs to be checked
  - Visualize the sampling chains (posterior samples form a chain)
  - Confirm that the potential scale reduction factor (PSRF) is < 1.05
- The posterior distribution can be used to estimate
  - Central tendency (mean, median, mode): most credible value
  - Variability (credible interval, highest density interval): credible range of values

Activity Time!

Let’s interpret results from a Bayesian CFA

Load the activity2.R file

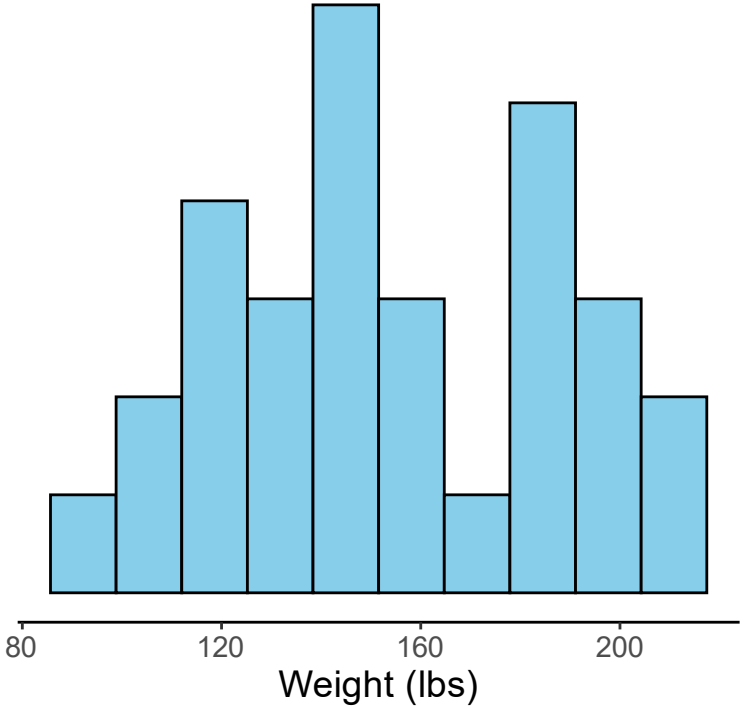


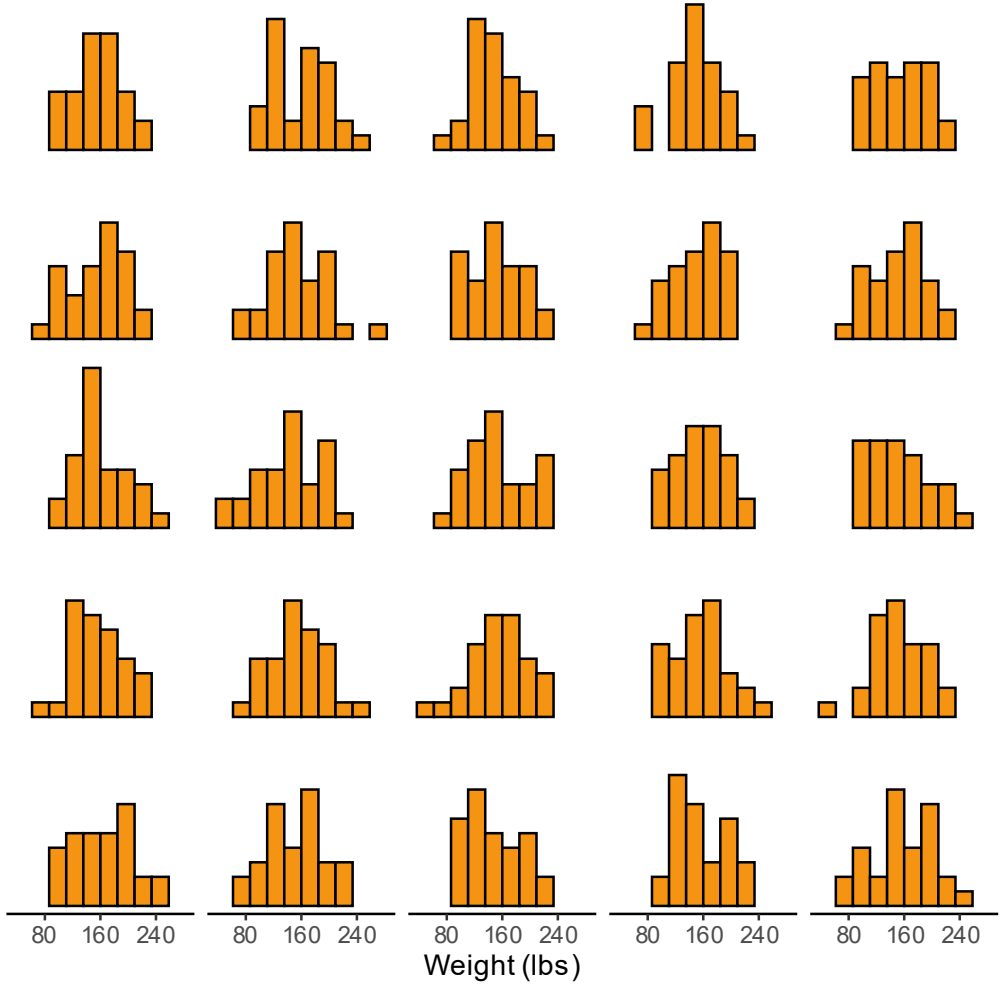


Bayesian Model Evaluation

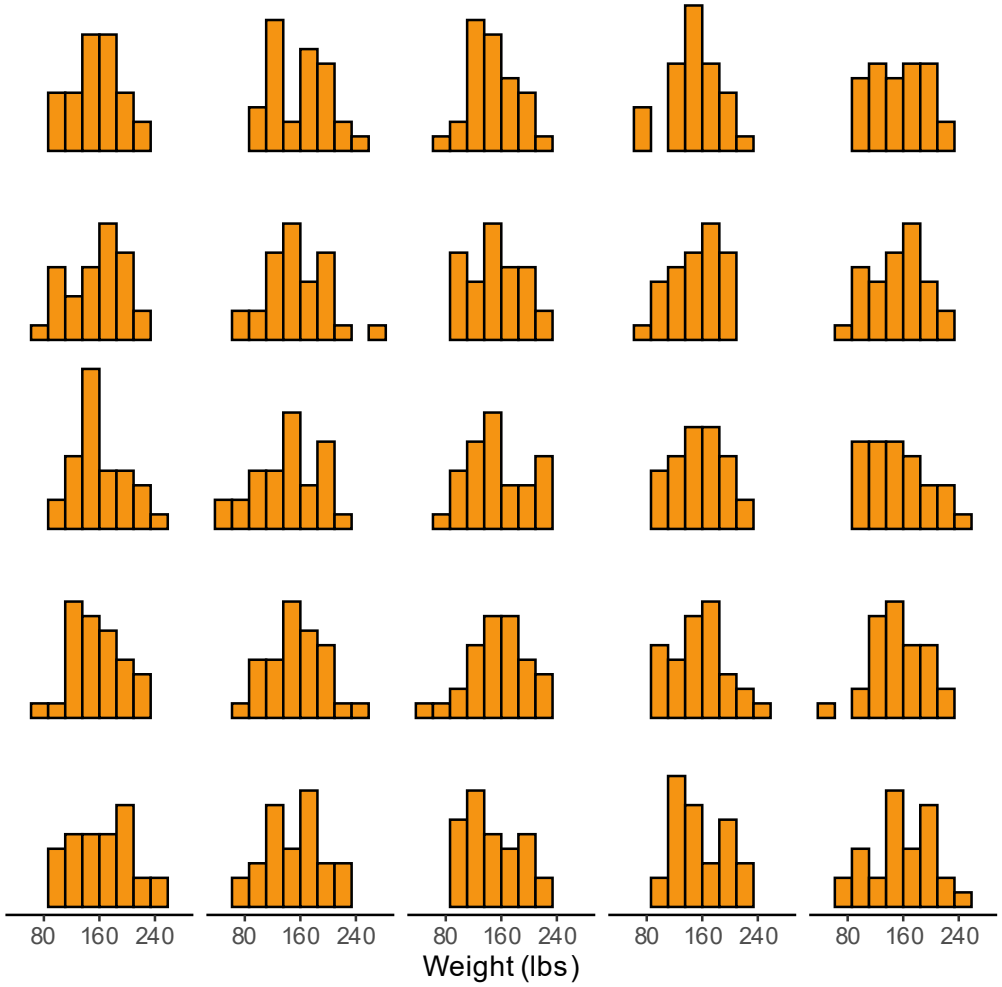
- Posterior distributions only tell us which parameter values are relatively less bad than others
  - They do not tell us whether the least bad parameter values are *actually* any good
- One way to evaluate whether the posterior distributions are any good is via *posterior predictive checks*
  - The idea: If the posterior distributions are a good description of the patterns observed in the data, then (fake) data predicted from those posterior distributions should *actually look like* the observed (real) data.

Is our Linear Regression Model any Good?

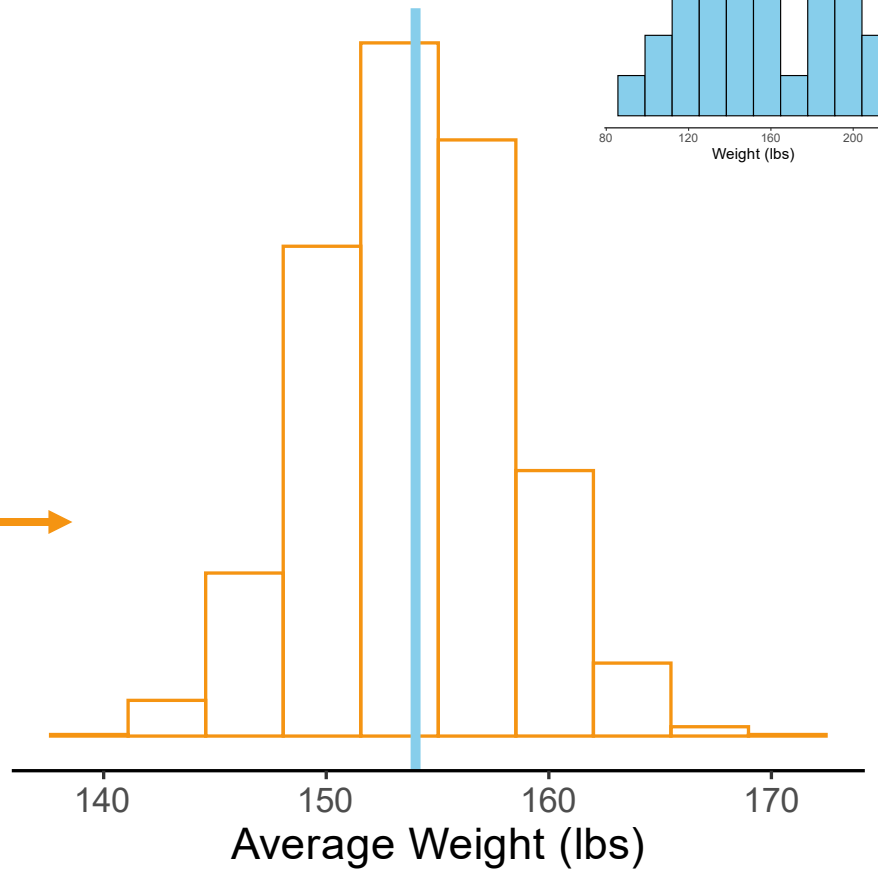




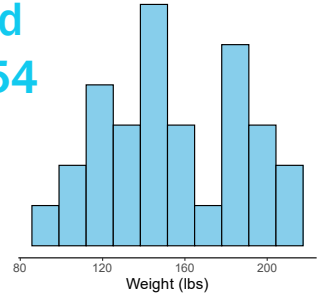
Looking at relevant test statistics



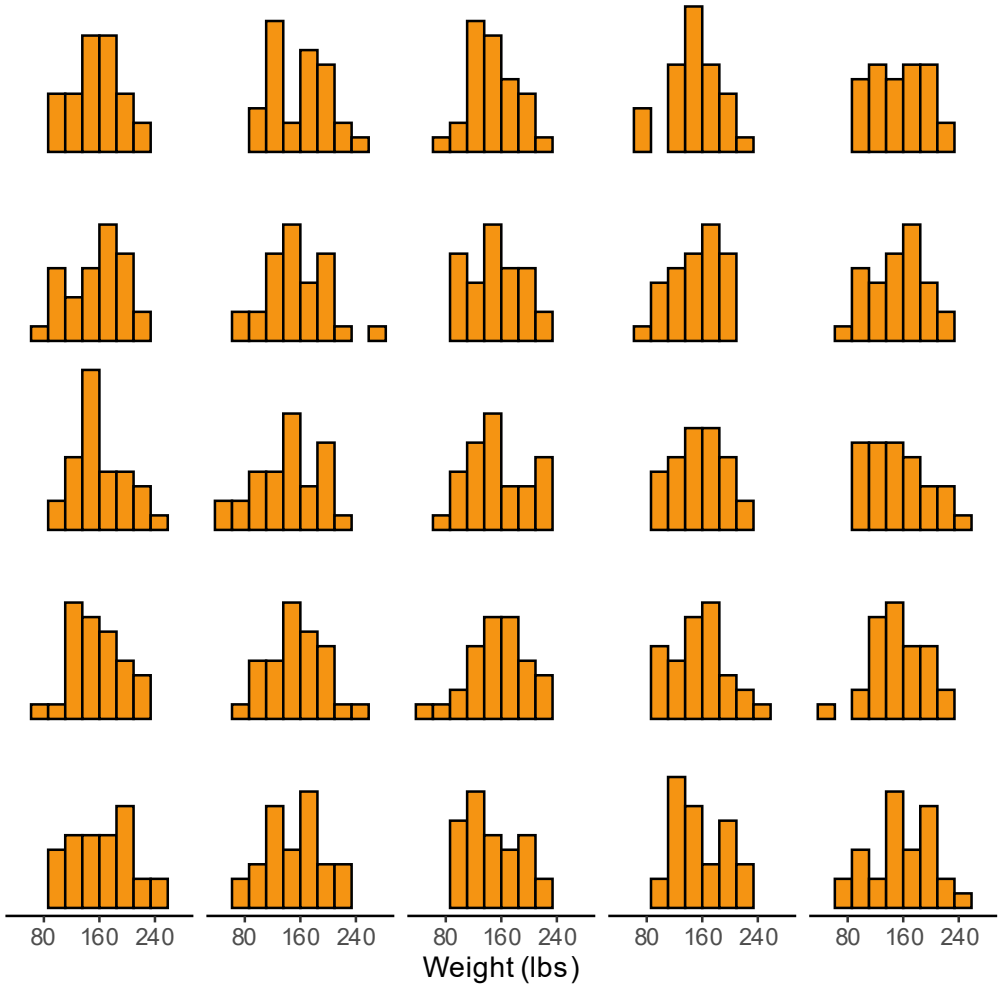
Mean Weight



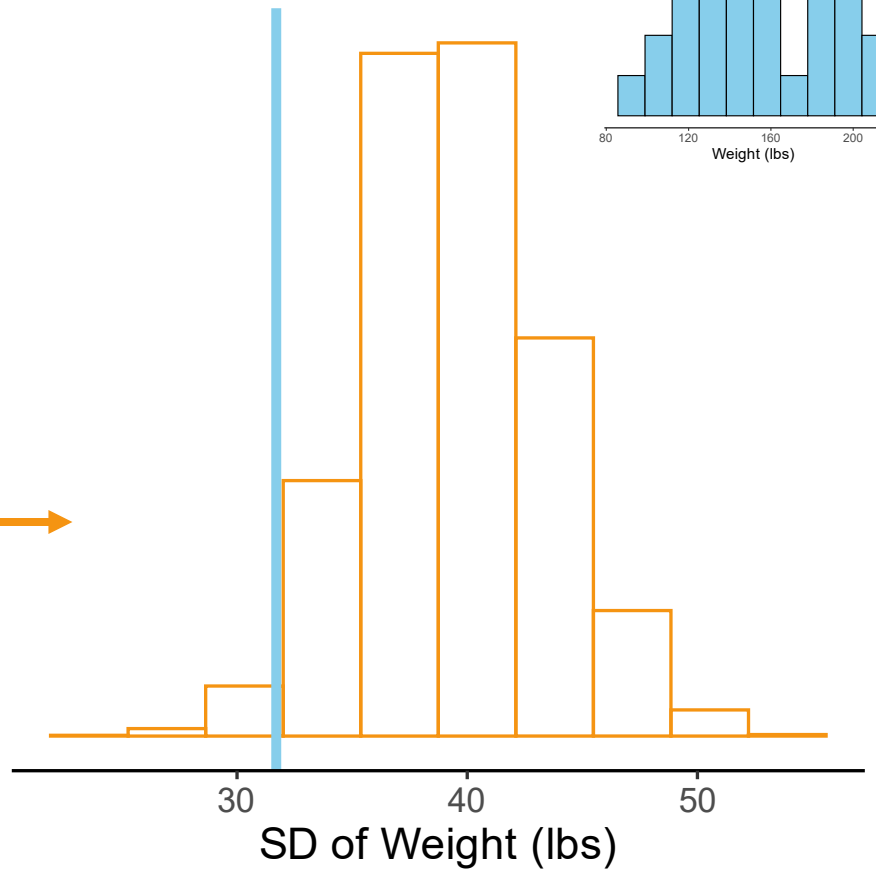
Observed
Mean: 154



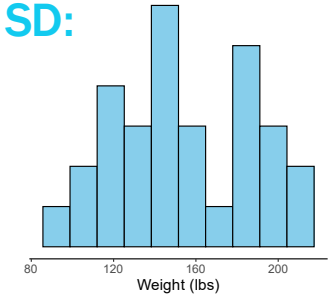
Looking at relevant test statistics



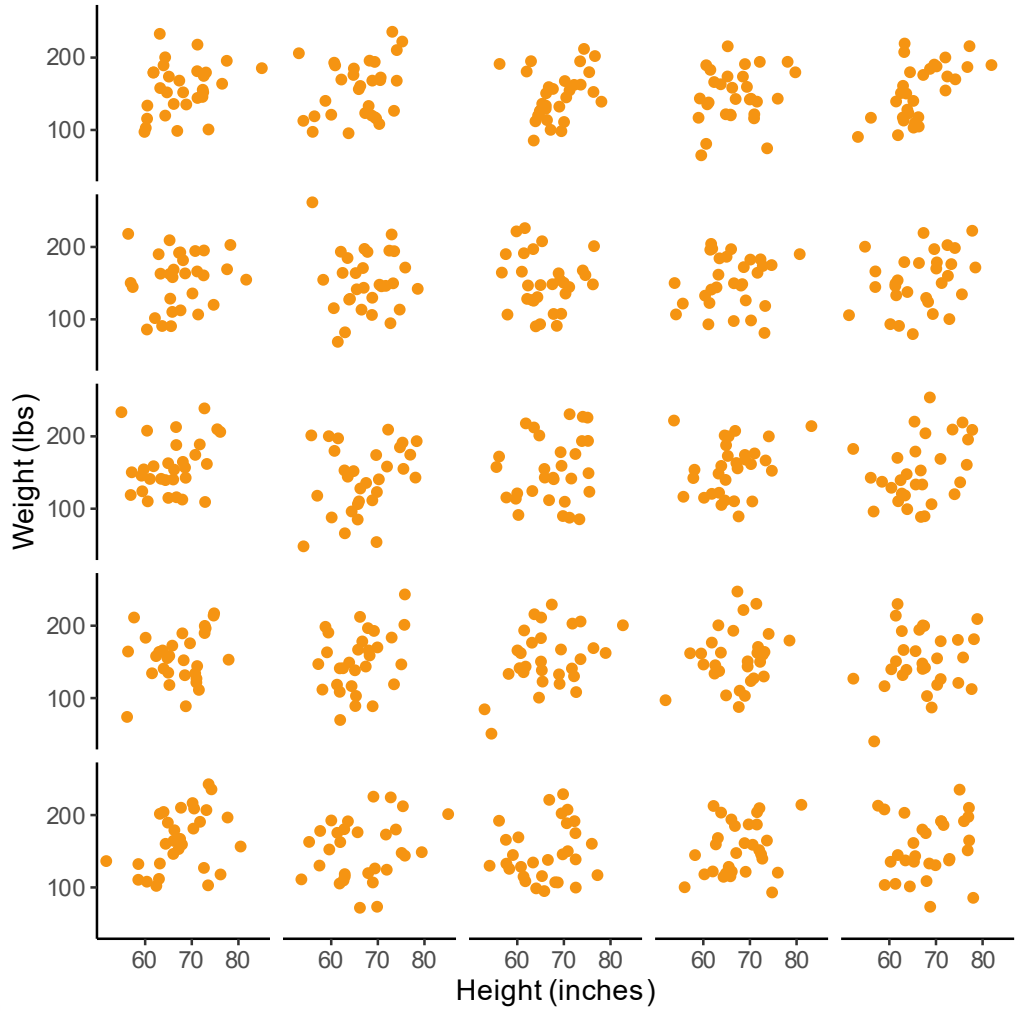
SD of Weight



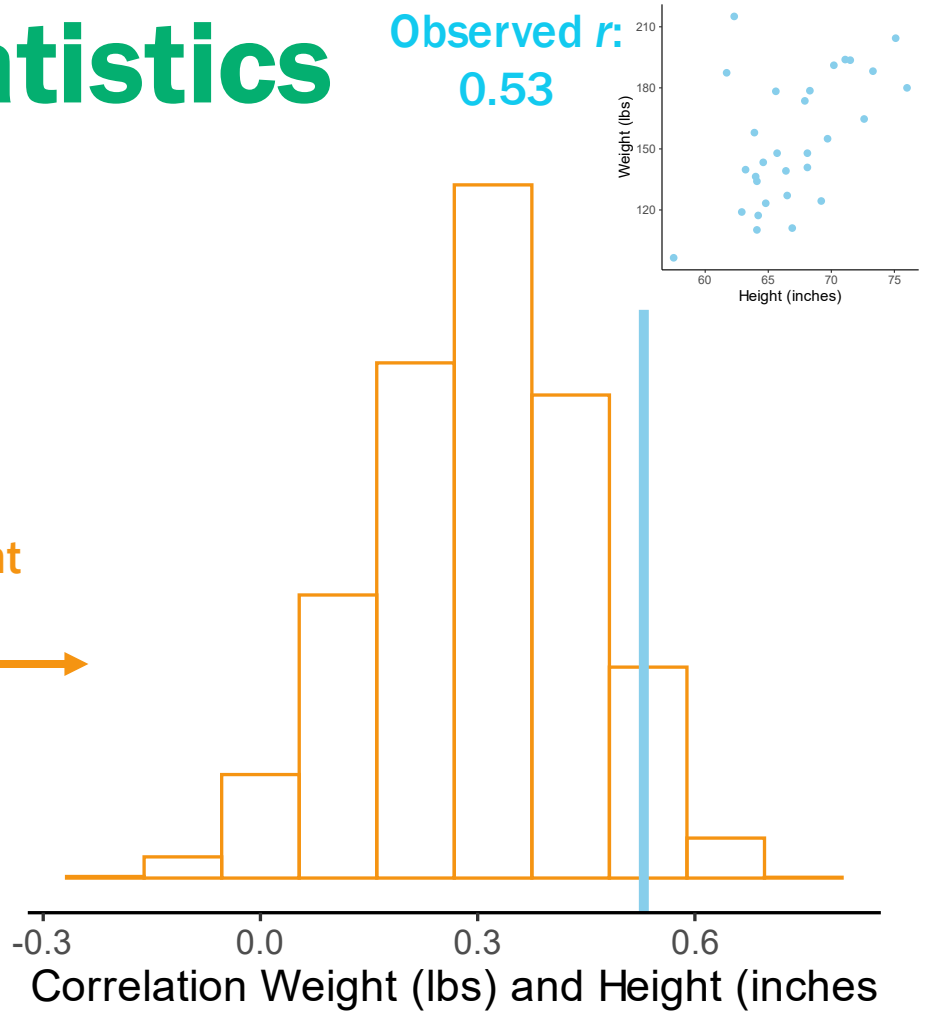
Observed SD: 31.7



Looking at relevant test statistics



Cor of Weight and Height



Bayesian Evaluation of CFAs

- Posterior predictive model checks can help evaluate the fit of a CFA based on different data characteristics
  - With CFA, can also look at model fit indices as test *quantities*, to evaluate if model fit is similar in predicted samples
- If 95% of predictive sample test statistics are above or below the sample estimate, it indicates an issue in at least one component:
  - Discrepancy between priors and new evidence: re-estimate model with different priors (sensitivity analysis)
  - Problem with data or (distributional) assumptions: select a model that aligns with data properties
  - Problem with model: re-specify model



Activity Time!

Let’s evaluate the fit of a Bayesian CFA using posterior predictive model checks

Load the activity3.R file

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Any questions about Bayesian model evaluation?

3. Information-Theoretic Model Evaluation



Example: Goodness-of-fit testing

Frequentist Statistics

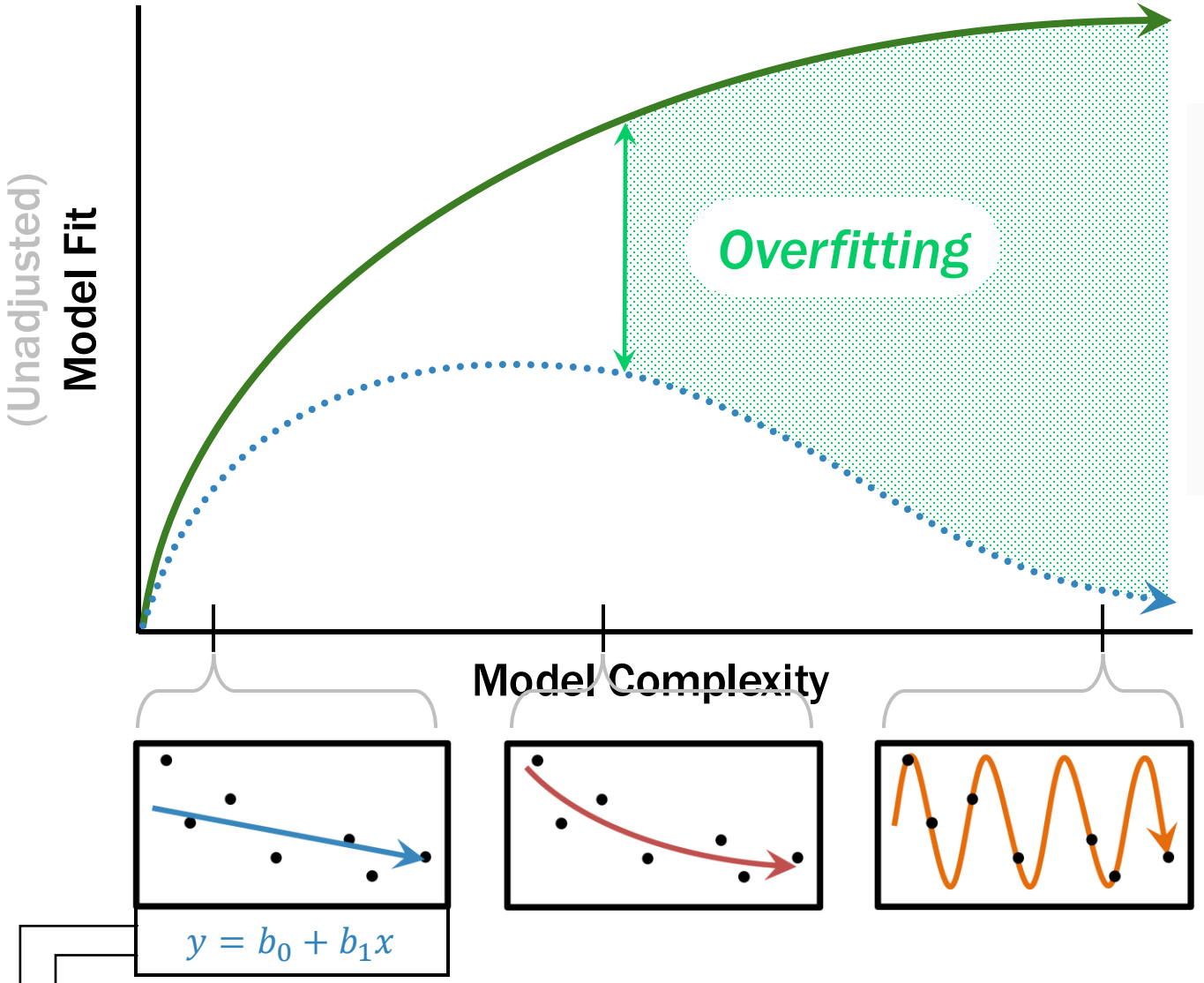
Example: Fitting propensity analysis

Information Theory

Bayesian Inference

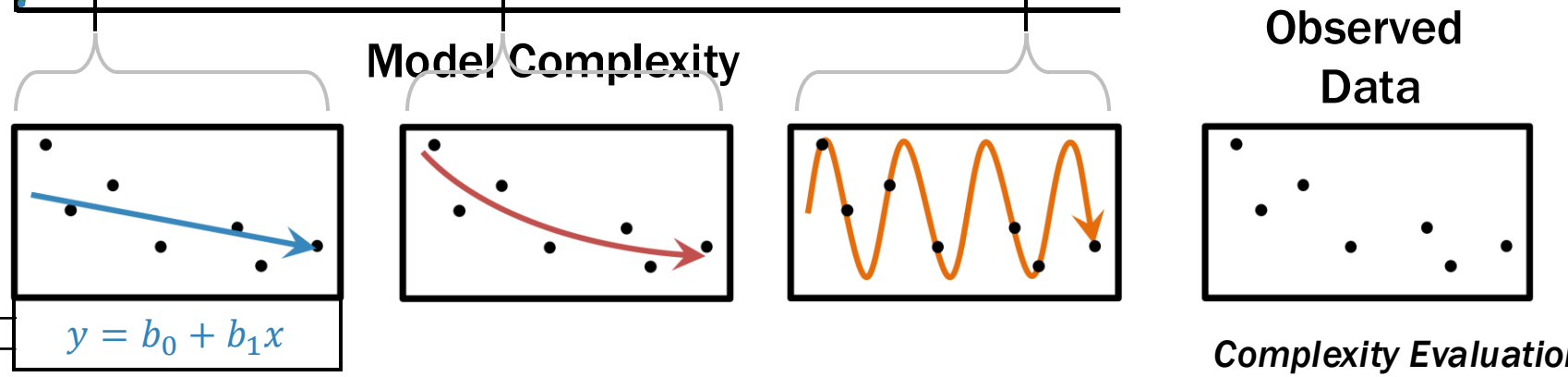
Example: Predictive model checking

(Adapted from Myung & Pitt, 2001)



— Goodness-of-fit of the model to the observed data

··· Generalizability of the model beyond the observed data



Complexity Evaluation:

More parameters: $y = b_0 + b_1 x + b_2 x^2$ $y = b_0 + b_1 x + \cdots + b_6 x^6$ e.g., AIC, BIC

Same # of parameters: $y = \frac{b_0}{e^{b_1 x}}$ $y = b_0 \cdot \sin(b_1 x)$ e.g., MDL, Fitting propensity

A method for investigating configural complexity

Myung, Pitt, & Kim (2005) provide a general definition of complexity:

"[A] model's inherent flexibility that enables it to fit a wide range of data patterns" (p. 12)

- To explore the role of model configuration in this inherent flexibility, we can generate a wide range of data patterns* and count how many patterns are fit well by models with the same number of parameters, but different configurations

* May as well consider *all possible* data patterns (i.e., the complete data space)

A method for investigating configural complexity

Our focus today will be on evaluating the configural complexity of factor analysis models

- For factor analysis, data take the form of correlation (or covariance) matrices
- Therefore we need to define the complete space of correlation matrices

Defining the complete data space

Consider a specific symmetric 3×3 matrix:

$$\mathbf{M}_k = \begin{bmatrix} 1 & & \\ .4 & 1 & \\ .1 & -.2 & 1 \end{bmatrix}$$

$$\mathbf{M}_k = \begin{bmatrix} 1 & & \\ x & 1 & \\ y & z & 1 \end{bmatrix}$$

$\mathbf{M}_k$ is just one of the ∞ points (e.g., 3×3 matrices) within this cube

Defining the complete data space

But not every symmetric matrix is a correlation matrix

- A symmetric matrix $\mathbf{M}$ is a correlation matrix $\mathbf{C}$ iff it is *positive semidefinite (PSD)*
  - i.e., its eigenvalues are non-negative, and its determinant ≥ 0

In the 3×3 case, we can label the unique elements of any $\mathbf{M}_k$ as $\mathbf{r} = (x, y, z)$

$$\mathbf{M}_k = \begin{bmatrix} 1 & & \\ x & 1 & \\ y & z & 1 \end{bmatrix}$$

- Within the $[-1,1]^3$ cube, each $\mathbf{r}$ is a single point, but only certain points qualify as $\mathbf{C}$s
- The boundary of the PSD subspace (i.e., the $\mathbf{C}$-space) is defined by all $\mathbf{r}$s that yield $\det(\mathbf{M}) = 0$
- This boundary results in the closed surface defined by

$$x^2 + y^2 + z^2 - 2xyz = 1$$

Defining the complete data space

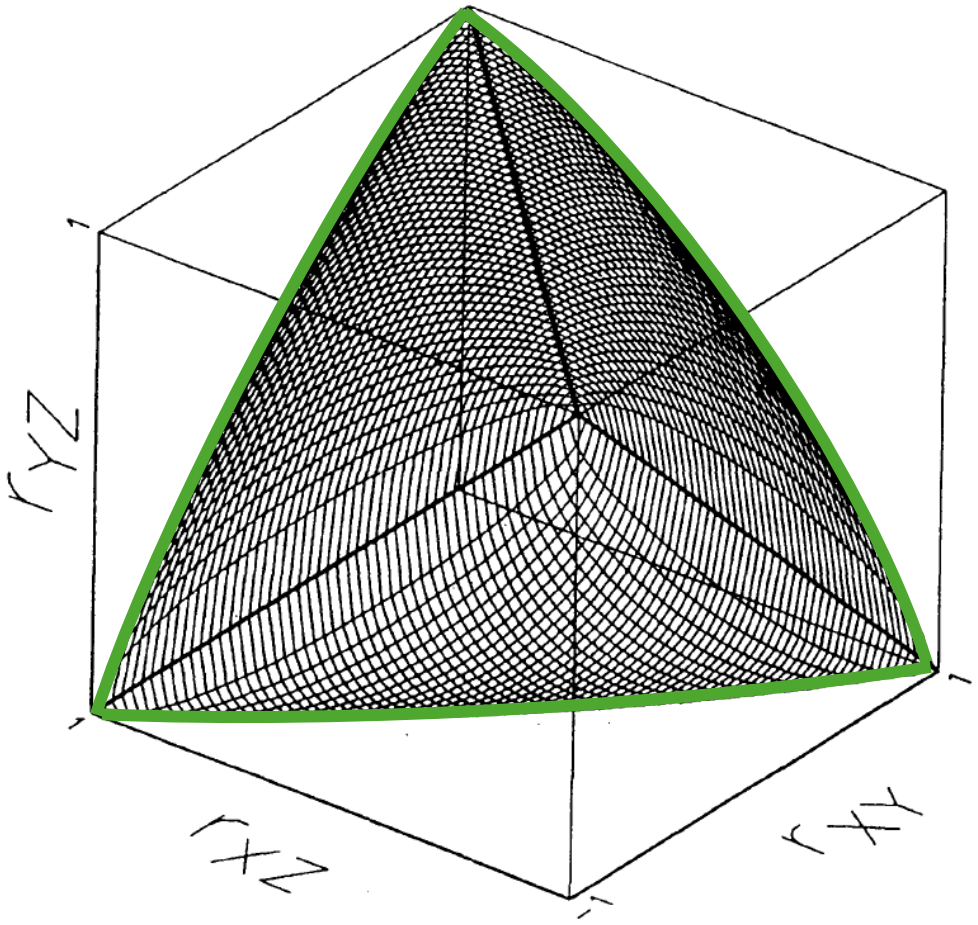


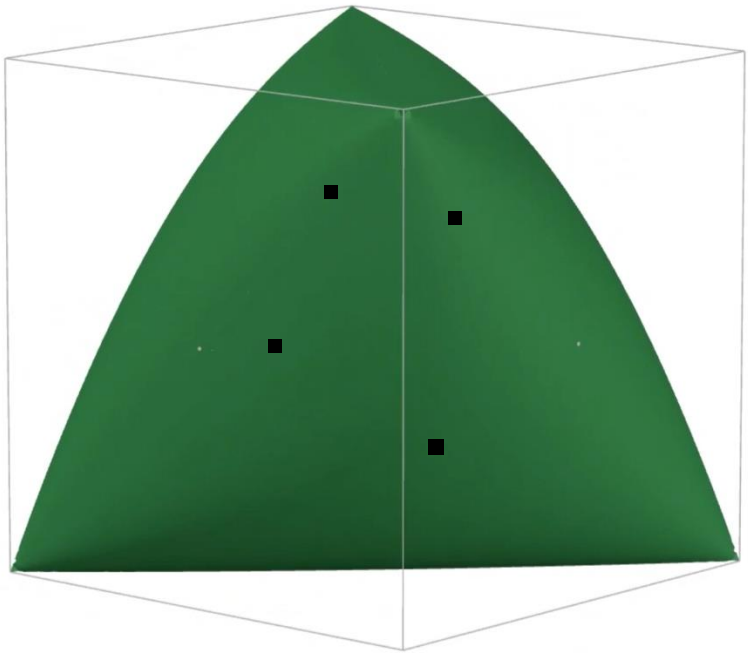
Figure from Rousseeuw & Molenberghs (1994). The shape of correlation matrices. *The American Statistician, 48*(4), 276-279.

- For 3×3 correlation matrices, the complete space forms an *elliptical tetrahedron*
- For $m \times m$ correlation matrices, where $m \geq 4$, the complete space is known as an *elliptope*

Evaluating model complexity

Step 1: Generate data by sampling many $\mathbf{C}_k$ matrices randomly and uniformly from the elliptope

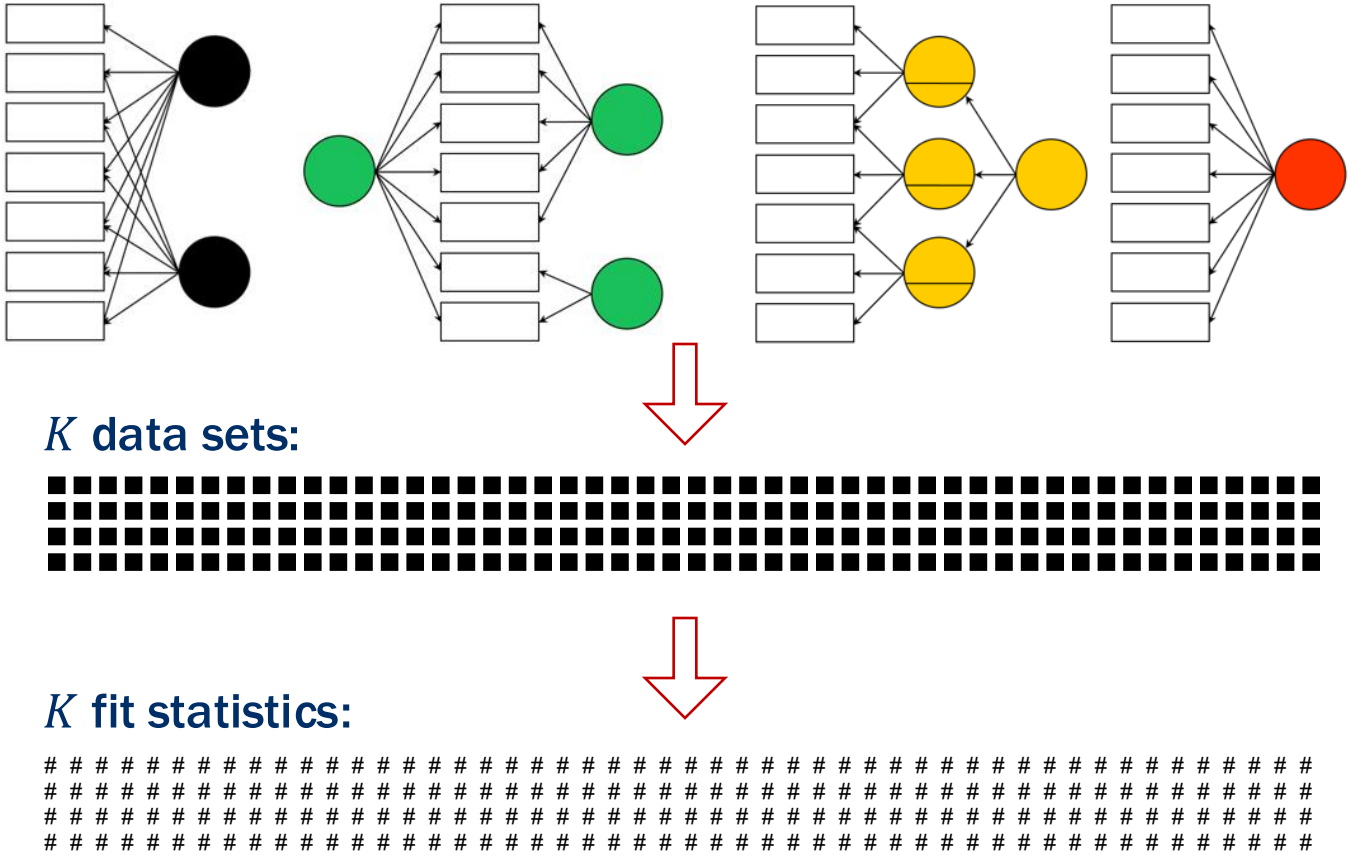
- where $k \in K$, and K is sufficiently large (e.g., 100,000 matrices)
- The *onion* sampling method (Lewandowski et al., 2009) ensures coverage of this space



$\mathbf{C}_1$
$\mathbf{C}_2$
$\mathbf{C}_3$
$\mathbf{C}_4$
... $\mathbf{C}_K$

Evaluating model complexity

Step 2: Fit all candidate models to the same set $\mathbf{C}_K$ to obtain K fit statistics



Evaluating model complexity

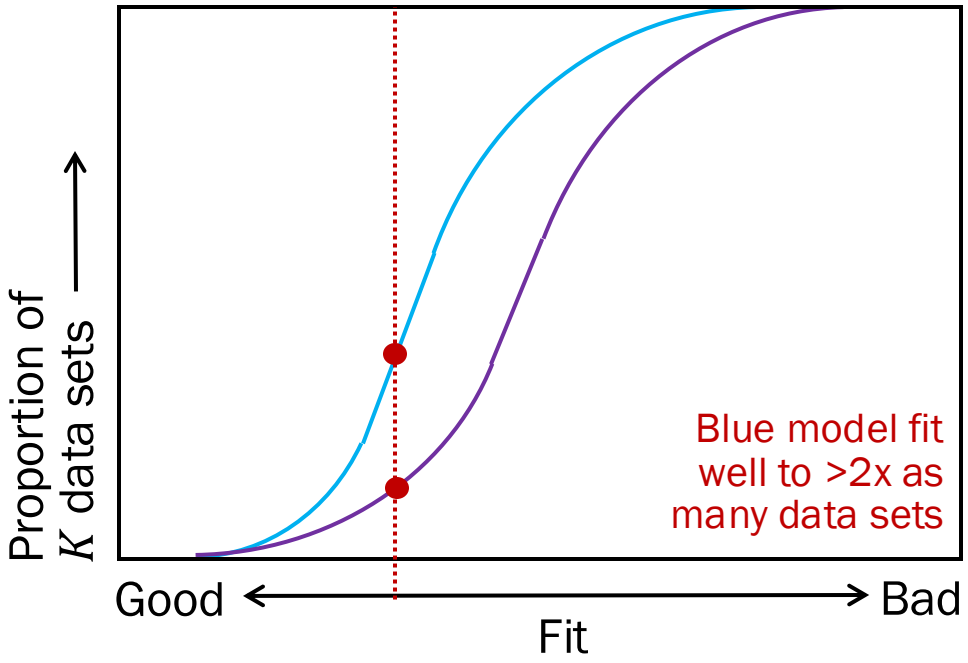
Step 3: Examine configural complexity by:

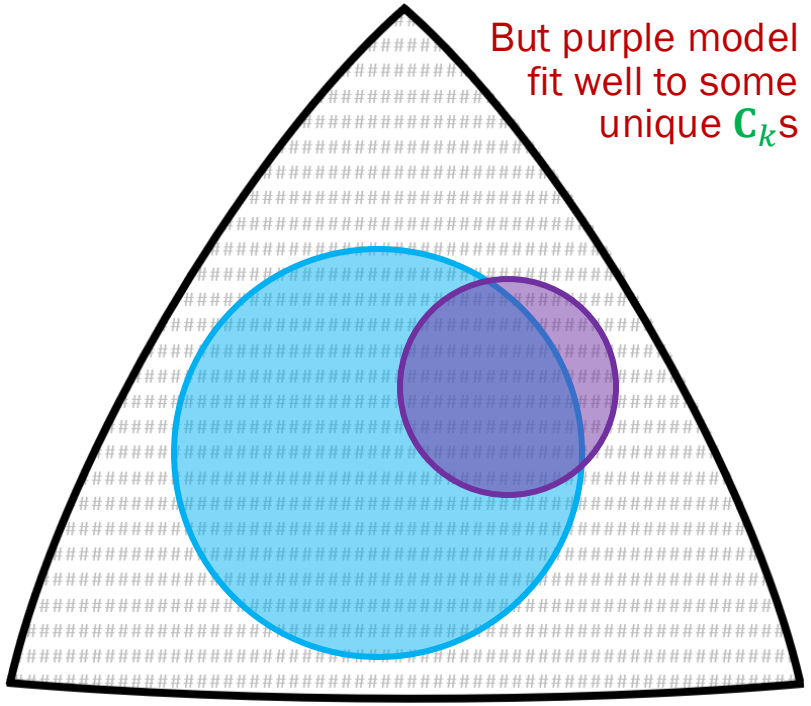
a) Counting how often a model achieves some goodness-of-fit threshold

- Left panel: Visualized as an empirical cumulative density function (ECDF)

b) Considering the unique and shared $\mathbf{C}_k$s accommodated by each model

- Right panel: Visualized as an Euler plot (area-proportional Venn diagram)





Normalized Maximum Likelihood

This technique aligns generally with the information-theoretic principle of *minimum description length* (Rissanen, 1978; Grünwald, 2004; Navarro, 2019)

- **More specifically, it resembles Rissanen's (2001) *normalized maximum likelihood:***

$$\mathrm{NML} = \frac{L(\mathrm{D}|\mathbf{M})}{\int_S L(\mathbb{D}|\mathbf{M}(\mathbb{D}))d\mathbb{D}}$$

- **where**
  - $L(\mathrm{D}|\mathbf{M})$: **the likelihood of observed data D given model M**
  - $L(\mathbb{D}|\mathbf{M}(\mathbb{D}))$: **the likelihood when the model is fit to all possible data $\mathbb{D}$ in the data space S**
- **NML quantifies the information in a model, adjusted for the model's propensity to fit well to *all possible data***
  - **Unfortunately, the denominator involves integration across the complete data space, so it is usually intractable**

“Quantifying parsimony in structural equation modeling”

In a NML-like simulation, Preacher (2006) examined the *fitting propensities (FPs)* of competing structural equation models that had the same number of parameters, but different configurations

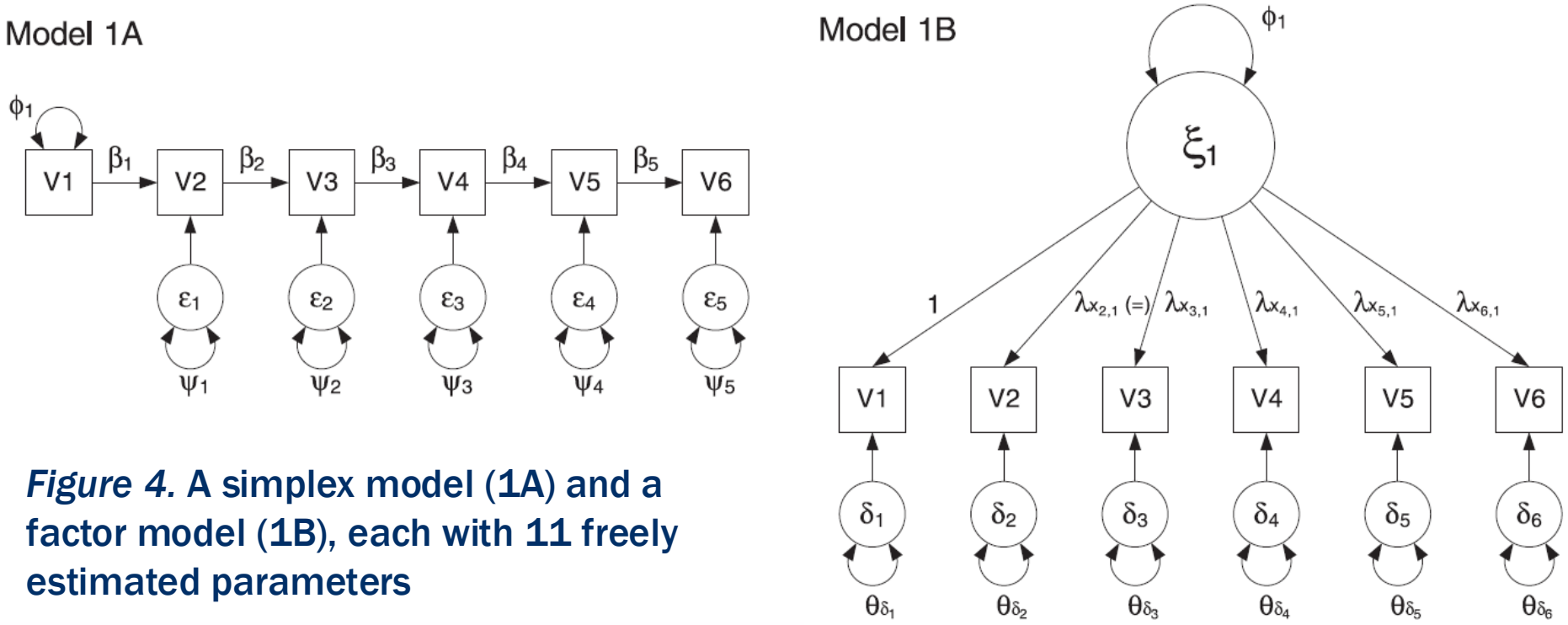


Figure 4. A simplex model (1A) and a factor model (1B), each with 11 freely estimated parameters

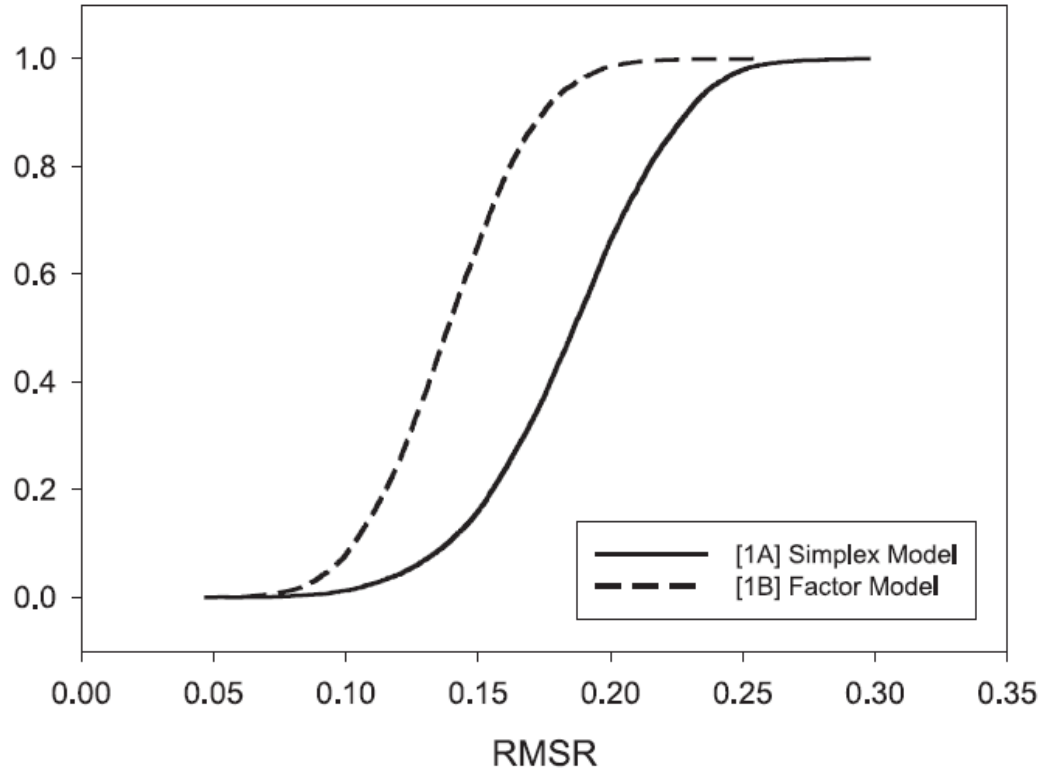
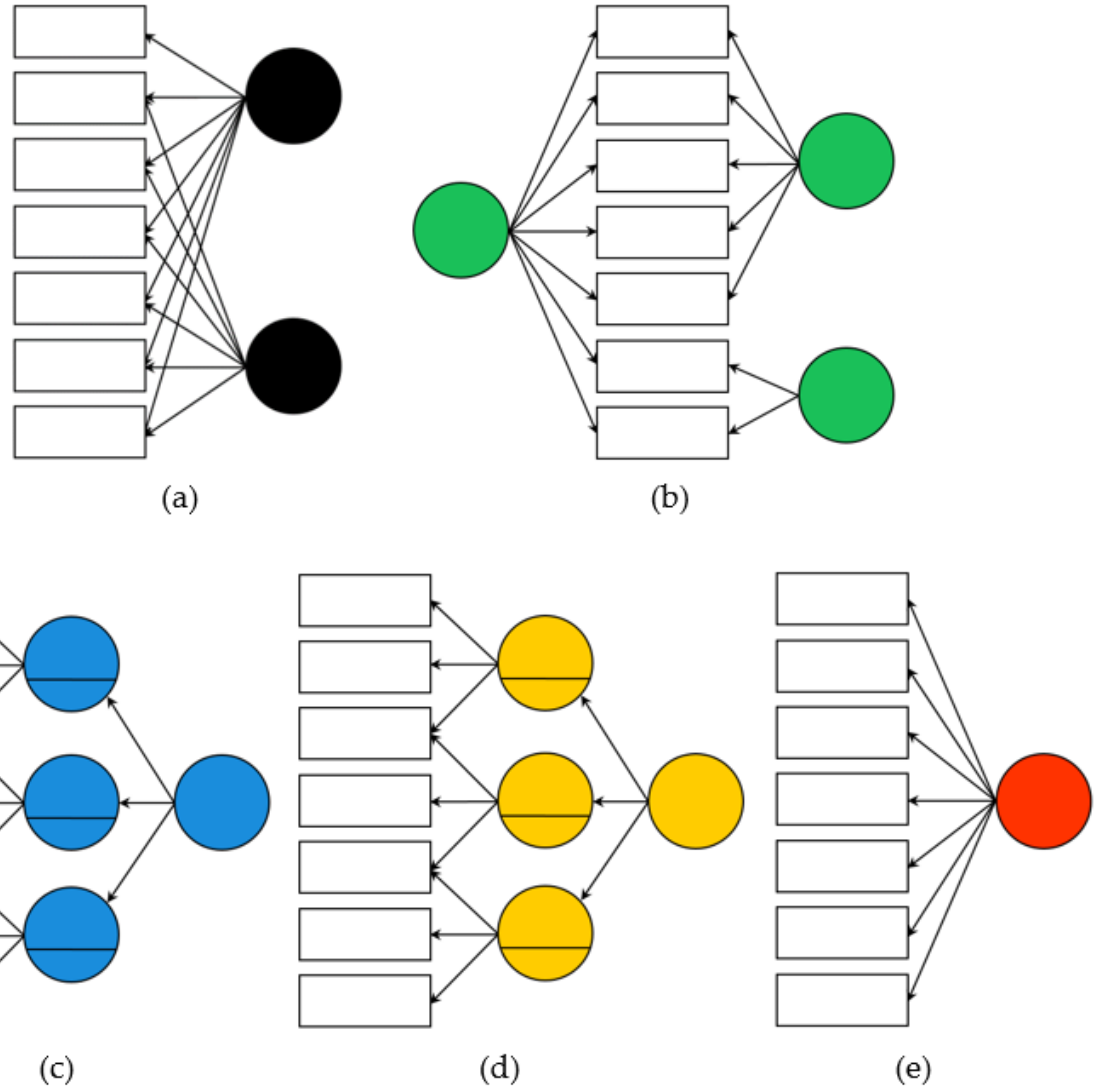


Figure 5. Cumulative frequency distributions of unadjusted goodness-of-fit (RMSR) when fitting Models 1A and 1B to the same 10,000 random correlation matrices

The common factor configuration (1B) has a higher propensity to fit well to any correlation matrix, *despite having the same number of parameters* as the simplex configuration (1A)

"The good fit of a hypothesized model to observed data, although desirable, can result from the model's inherent ability to predict data patterns and may have little to do with its value as a scientific tool. **Cherished models may have to be abandoned or replaced** if their past successes can be ascribed more to FP than to any insight they lend into the process that actually generated the data." (p. 254)

Bonifay & Cai (2017) extended Preacher (2006) to the categorical case (item factor analysis)



5 measurement models:

(a) EIFA

(b) Bifactor

(c) DINA

(d) DINO

20 free parameters

(e) Unidimensional (3PL)

21 free parameters

Fit to 1,000 data sets & evaluated via the Y2/N unadjusted fit index

EFA

- 7 c (intercept) parameters
- 7 a (discrimination) parameters for Factor 1
- + 6 a (discrimination) parameters for Factor 2
- **20 free parameters**

Bifactor

- 7 c (intercept) parameters
- 7 a (discrimination) parameters for the General Factor
- 5 a (discrimination) parameters for Specific Factor 1
- + 1 a (discrimination) parameter for Specific Factor 2
- **20 free parameters**

DINA & DINO

- 7 $\lambda_{1,0}$ (intercept) parameters
- 2 $\lambda_{1,1,(1)}$ (main effect) parameters for Attribute 1
- 1 $\lambda_{1,1,(2)}$ (main effect) parameter for Attribute 2
- 2 $\lambda_{1,1,(3)}$ (main effect) parameters for Attribute 3
- 1 $\lambda_{1,2,(1,2)}$ (interaction effect) parameter for Attributes 1 & 2
- 1 $\lambda_{1,2,(2,3)}$ (interaction effect) parameter for Attributes 2 & 3
- 3 c (attribute intercept) parameters
- + 3 a (attribute discrimination) parameters
- **20 free parameters**

Uni

- 7 c (item intercept) parameters
- 7 a (item discrimination) parameters
- + 7 g (pseudo-guessing) parameters
- **21 free parameters**

Categorical Data Generation

Data generation:

Sampling from a unit simplex *(Smith & Tromble, 2004)*

- Generate all possible item response patterns for a small number of binary response items

  ↳ $2^7 = 128$ possible patterns

- Assign to each a random weight representing the number of simulees (of $N = 10{,}000$) who supplied that particular pattern

1,000 random data sets to represent the complete data space

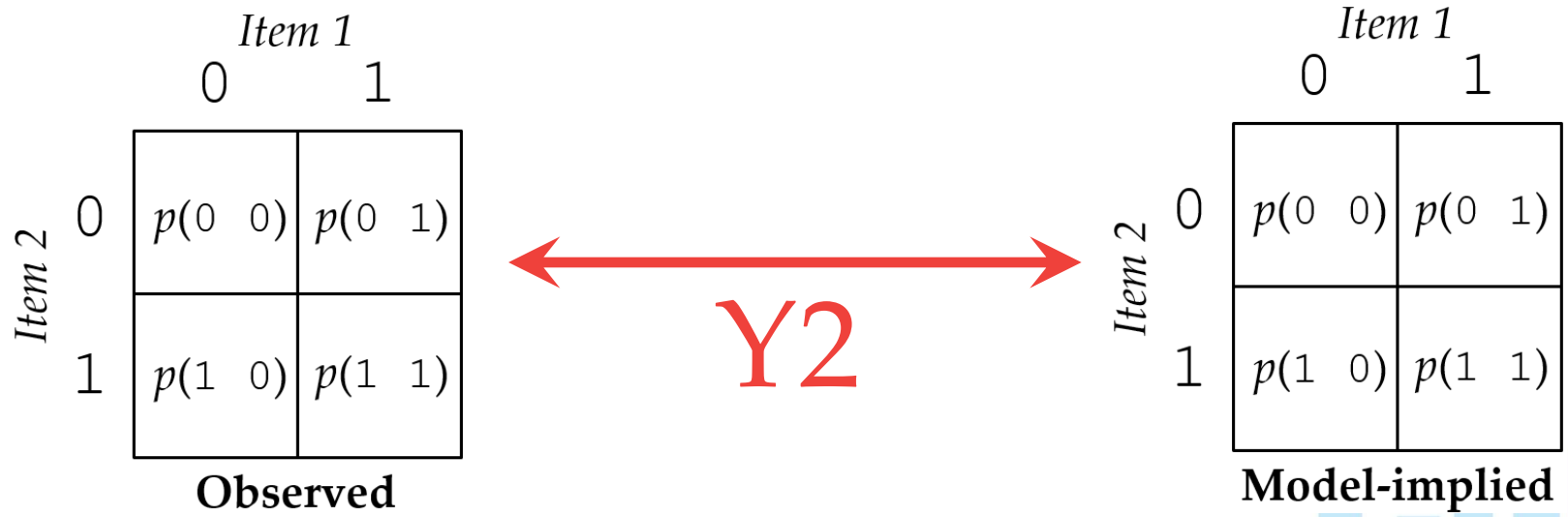
Example data set:

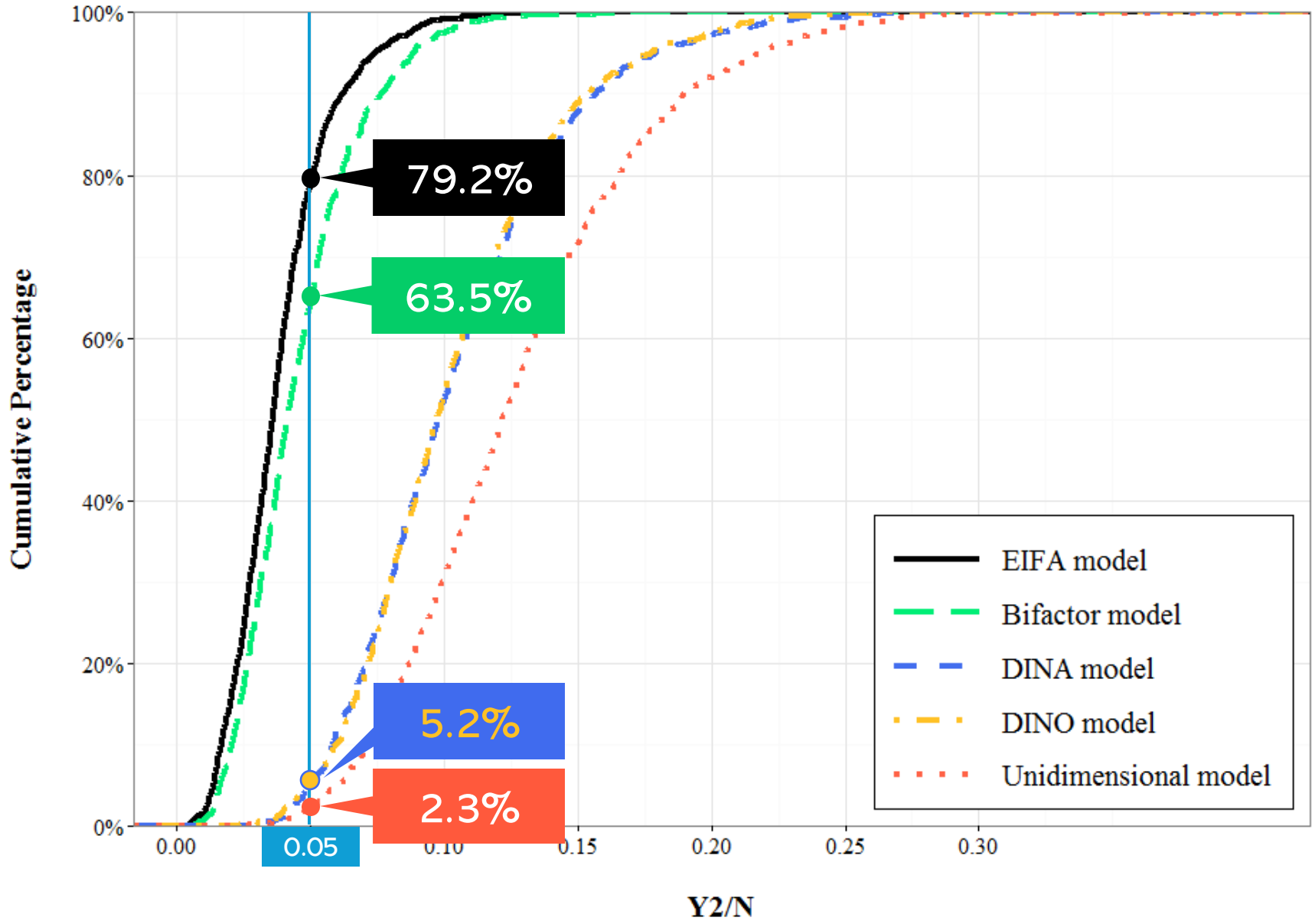
```
0 0 0 0 0 0 0 20.348
0 0 0 0 0 0 1 60.367
0 0 0 0 0 1 0 27.676
0 0 0 0 0 1 1 7.983
0 0 0 0 1 0 0 3.714
0 0 0 0 1 0 1 10.517
0 0 0 0 1 1 0 47.091
          ⋮
1 1 1 1 0 0 1 4.961
1 1 1 1 0 1 0 30.693
1 1 1 1 0 1 1 14.663
1 1 1 1 1 0 0 1.993
1 1 1 1 1 0 1 2.673
1 1 1 1 1 1 0 67.551
1 1 1 1 1 1 1 40.919
```

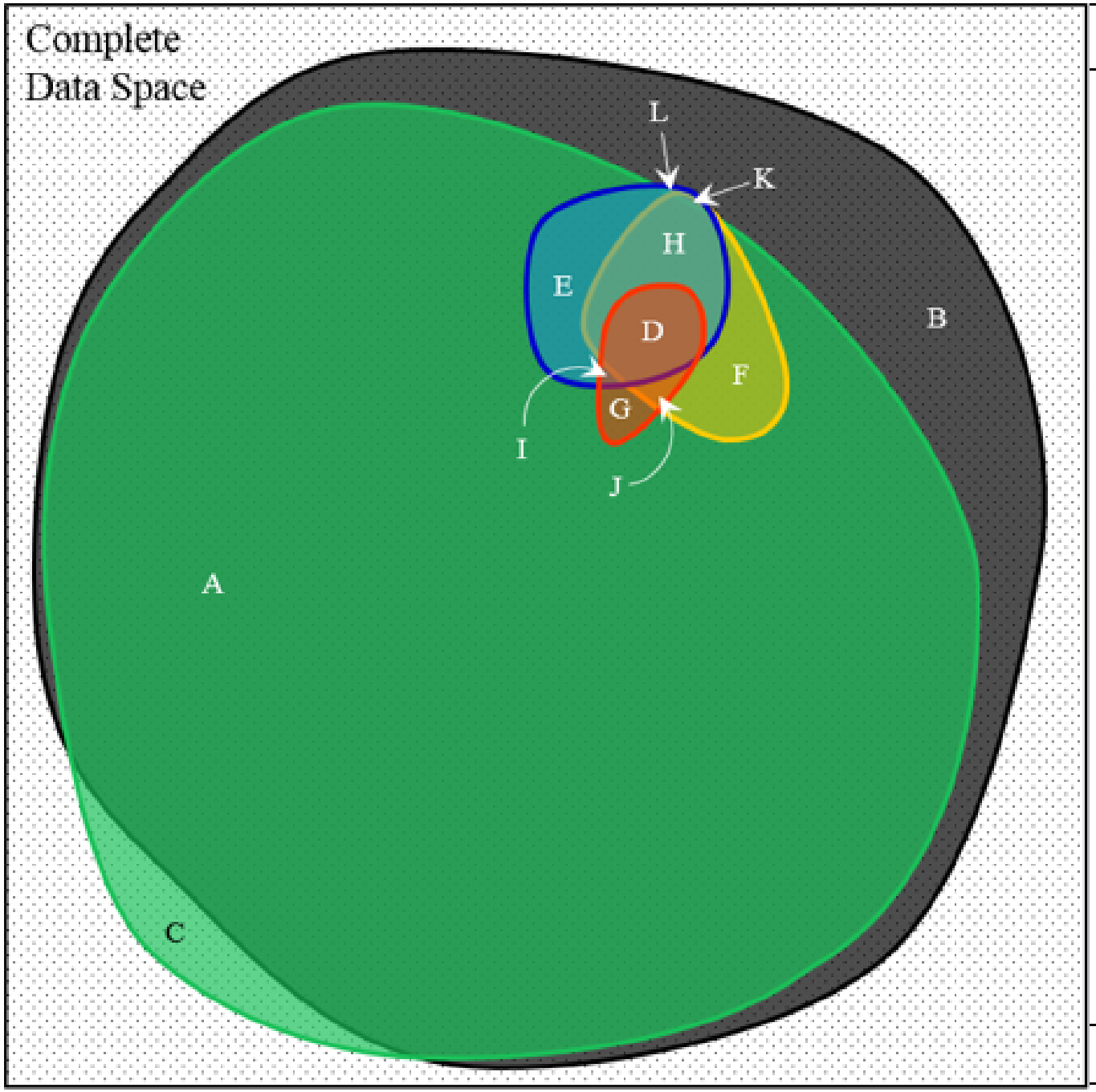
Evaluation criterion

- **Y2/*N* statistic** *(Bartholomew & Leung, 2002; Cai et al., 2006)*:

$$Y2/N = \left[\sum_{i=1}^{I} \frac{(o_i - e_i)^2}{e_i(1-e_i)} + \sum_{i=1}^{I-1} \sum_{j=i+1}^{I} \frac{(o_{ij} - e_{ij})^2}{e_{ij}(1-e_{ij})}\right]$$




100%
80%
60%
40%
20%
0%
Cumulative Percentage
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Y2/N
79.2%
63.5%
5.2%
2.3%
EIFA model
Bifactor model
DINA model
DINO model
Unidimensional model



Region	Y2/N ≤ .05
EIFA	79.2 %
Bifactor	63.5
DINA	5.1
DINO	5.2
Uni	2.3
A	51.8
B	19.2
C	3.8
D	1.1
E	1.9
F	2.0
G	0.7
H	1.7
I	0.2
J	0.3
K	0.1
L	0.1
Unoccupied	17.1

Bonifay & Cai (2017)

Results reiterate the importance of *configural* complexity:

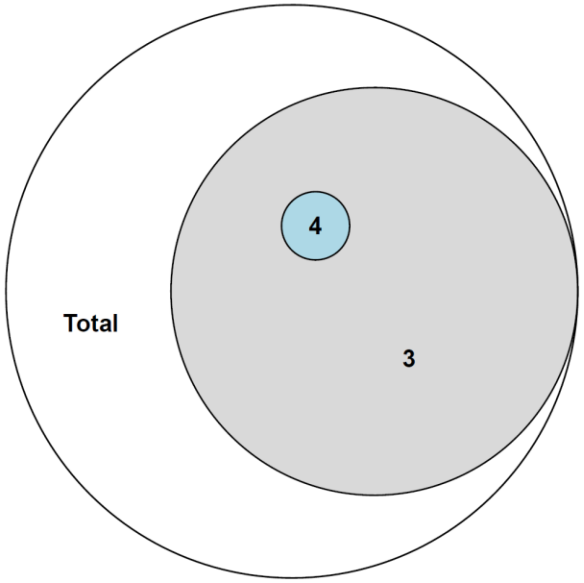
- The configuration of the variables in a model affects its propensity to fit well
- More configurally complex models (EIFA & bifactor) displayed high propensity to fit *any possible* data
- DINA & DINO diagnostic classification models had low FP
  - There is a distinct theoretical difference between these models, so they occupied different regions of the data space
- The unidimensional model had ***an additional free parameter***, but much lower FP!
- Strong implications re: model evaluation via goodness-of-fit
  - For models with high FP, overfitting is a statistical inevitability
  - For models with low FP, good fit is unlikely and thus far more meaningful when obtained

Falk & Muthukrishna (2021)

Extends Preacher (2006) by making FP evaluation (of continuous data models) more accessible

Provides a detailed tutorial of their `ockhamSEM` package in R

- Takes `lavaan` code as model specification input
- Generates *n* random correlation matrices
  - Improves upon Preacher's method by (a) using a much more efficient method of generating random correlation matrices, and (b) taking advantage of parallel computing
- Fits the model(s) to all *n* matrices
- Summarizes FP via text output, data visualization, & measures of effect size



Euler plot comparing Models 3 and 4 from Falk & Muthukrishna (2021)

Activity Time!

Let’s compare the fitting propensities of two CFA models

Load the activity4.R file





Any questions about information-theoretic model evaluation?

4. Summary



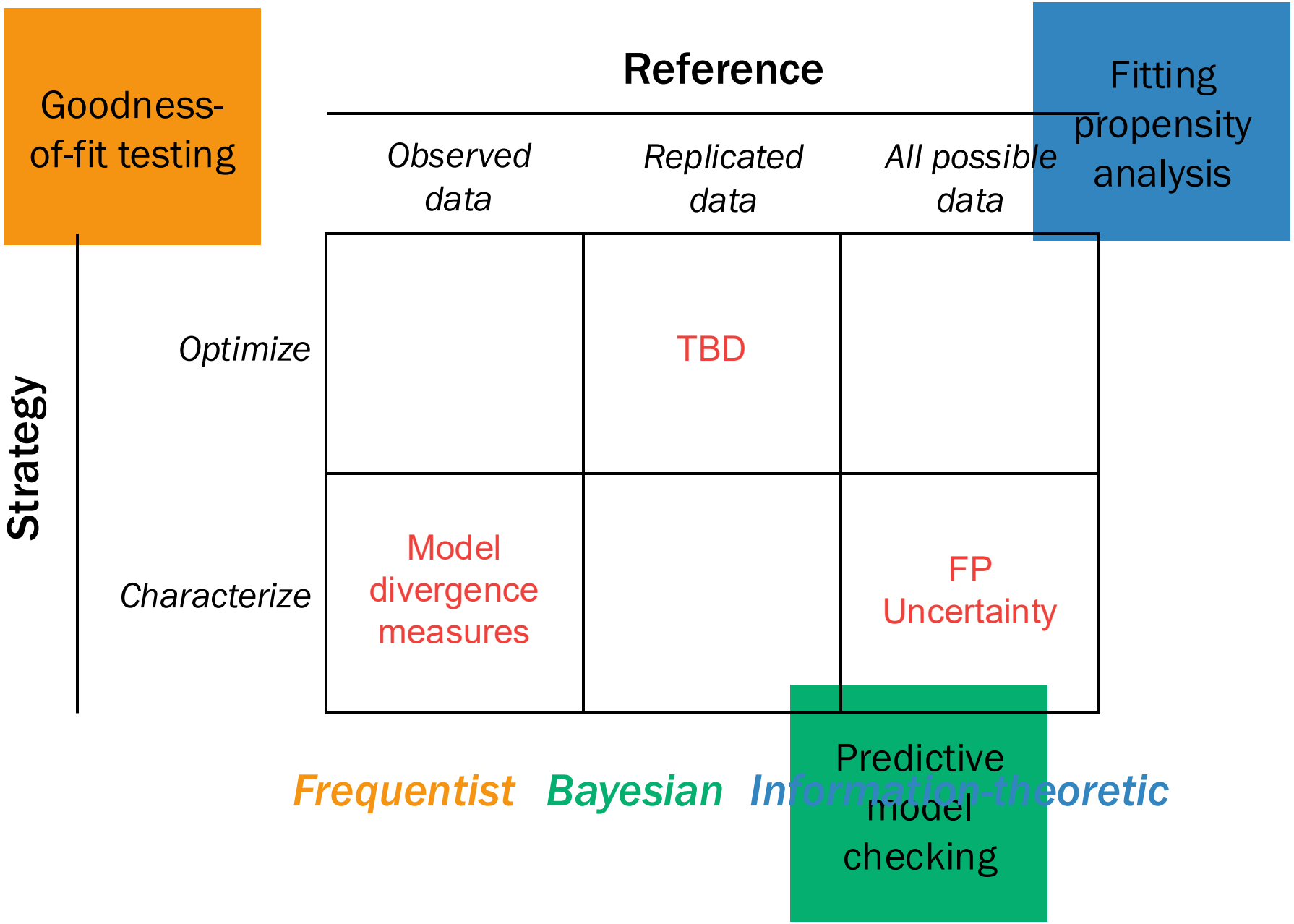
Summary of the Three Perspectives

Philosophical differences (where $\in$ means "is a member of the set" and $\mathbb{D}$ represents all possible data):

- **Frequentist:** $p(D \in \mathbb{D}|M)$, the probability of the *particular* observed data given the model
- **Bayesian:** $p(M|D \in \mathbb{D})$, the probability of the model given the *particular* observed data
- **Information-theoretic:** $p(M|\mathbb{D})$, the probability of the model given the *complete* data space

Methodological differences:

- Treatment of the data (observed, replicated, or all possible)
- Goal of evaluation (optimize to some point estimate or characterize uncertainty)


Goodness-of-fit testing
Reference
Fitting propensity analysis
Observed data
Replicated data
All possible data
Strategy
Optimize
Characterize
TBD
Model divergence measures
FP Uncertainty
Predictive model checking
Frequentist
Bayesian
Information-theoretic

A Framework for Comprehensive Statistical Model Evaluation

- Unites three seemingly disparate statistical philosophies, each with their own goals and techniques
- Can be used as a blueprint for model evaluation, thereby guiding researchers in making more comprehensive model evaluations
- Will lead to more informed decision-making and facilitate more rigorous evaluation of the theories underlying our models

5. Software



`cosme` Package in R

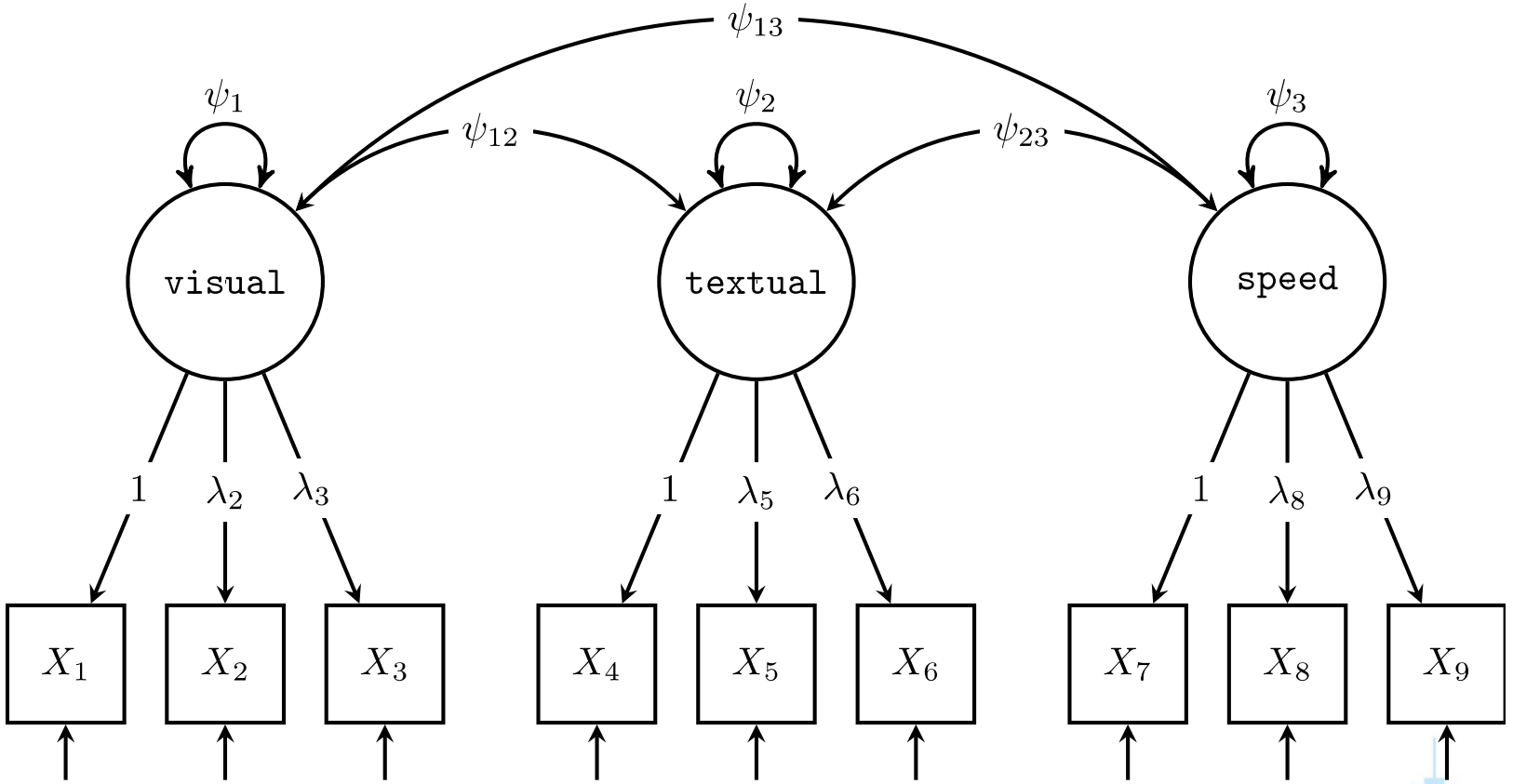
- Still under development and available at https://github.com/bkeller2/cosme

- Will be available on CRAN upon completion

Data Example: `HolzingerSwineford1939`

- A 'classical' data set is often used by SEM books and papers.
- The dataset contains mental ability test scores of two schools' seventh- and eighth-grade children.
- For 9 of the original 26 tests, a three-factor CFA model, each with three indicators, is often proposed.

Data Example: Path Diagram



R Code: Installing Package

```r
## Install package from github
remotes::install_github('bkeller2/cosme')

## Load Package
library(cosme)

## Get Holzinger Data from `lavaan`
data(HolzingerSwineford1939, package = 'lavaan')
HSdata <- HolzingerSwineford1939[ ,7:15]
```

R Code: Specifying the Model

```
## Set up model via `lavaan` syntax
## Uses `lavaan::cfa()` which assumes factors
##   are correlated
m <- c(
    "visual  =~ x1 + x2 + x3",
    "textual =~ x4 + x5 + x6",
    "speed   =~ x7 + x8 + x9"
)
```

R Code: Estimating All Models

```
## Runs 10000 reps for Information Theory
o <- cosme(
    m,              # Model Specified
    HSdata,         # Data Set
    info = TRUE     # Setting to TRUE for Info
)
```

Waiting...

```
→ Fitting Frequentist
→ Fitting Bayesian
→ Fitting Information Theory
✔ Matrices Generated.
→ Fitting Models to Matrices.
  Fitting ████████████████████████████████  10000/10000 | Elapsed: 23m 49.9s| ETA:  0s
✔ Models fit to all matrices.
```

R Code: Summarizing Results

```
## There will be 3 slots in `o`
# - `info`  information theory results
# - `freq`  frequentist results
# - `bayes` bayesian results

## Print out summaries
estimates(o)                # Estimates
estimates(o, std = TRUE) # Standardized Estimates
fit(o)                      # Fit statistics
```

Output: Estimates of Loadings

```
-- Latent Variables:
                   Estimate          Lower Bound (2.5%)    Upper Bound (97.5%)
                   Freq      Bayes |      Freq      Bayes |      Freq      Bayes
  visual =~                        |                       |
    x1             1.000     1.000 |                       |
    x2             0.554     0.575 |     0.358     0.356 |     0.749     0.832
    x3             0.729     0.759 |     0.516     0.527 |     0.943     1.029
  textual =~                       |                       |
    x4             1.000     1.000 |                       |
    x5             1.113     1.123 |     0.985     0.994 |     1.241     1.265
    x6             0.926     0.933 |     0.817     0.823 |     1.035     1.049
  speed =~                         |                       |
    x7             1.000     1.000 |                       |
    x8             1.180     1.239 |     0.857     0.938 |     1.503     1.632
    x9             1.082     1.194 |     0.785     0.801 |     1.378     1.888
```

Output: Estimates of Covariances

```
-- Covariances:
                   Estimate          Lower Bound (2.5%)    Upper Bound (97.5%)
                   Freq      Bayes |      Freq      Bayes |      Freq      Bayes
  visual ~~                        |                      |
    textual       0.408      0.389 |     0.264      0.243 |     0.552      0.548
    speed         0.262      0.243 |     0.152      0.145 |     0.373      0.355
  textual ~~                       |                      |
    speed         0.173      0.162 |     0.077      0.076 |     0.270      0.265
```

Output: Estimates of Variances

```
-- Variances:
                   Estimate            Lower Bound (2.5%)    Upper Bound (97.5%)
                    Freq      Bayes |      Freq      Bayes |      Freq      Bayes
   .x1             0.549      0.574 |     0.326      0.313 |     0.772      0.821
   .x2             1.134      1.141 |     0.934      0.943 |     1.333      1.376
   .x3             0.844      0.847 |     0.667      0.656 |     1.022      1.054
   .x4             0.371      0.380 |     0.278      0.286 |     0.465      0.484
   .x5             0.446      0.452 |     0.332      0.348 |     0.561      0.573
   .x6             0.356      0.361 |     0.272      0.277 |     0.441      0.454
   .x7             0.799      0.833 |     0.640      0.660 |     0.959      1.032
   .x8             0.488      0.510 |     0.342      0.317 |     0.633      0.707
   .x9             0.566      0.551 |     0.427      0.340 |     0.705      0.738
    visual         0.809      0.790 |     0.524      0.513 |     1.094      1.118
    textual        0.979      0.979 |     0.760      0.776 |     1.199      1.216
    speed          0.384      0.354 |     0.215      0.172 |     0.553      0.554
```

Output: Standardized Estimates of Loadings

```
-- Latent Variables:
                   Estimate          Lower Bound (2.5%)    Upper Bound (97.5%)
                     Freq      Bayes |      Freq      Bayes |      Freq      Bayes
  visual =~                          |                       |
    x1              0.772      0.761 |     0.664      0.633 |     0.880      0.875
    x2              0.424      0.432 |     0.307      0.292 |     0.540      0.546
    x3              0.581      0.591 |     0.473      0.465 |     0.689      0.703
  textual =~                         |                       |
    x4              0.852      0.849 |     0.807      0.800 |     0.896      0.894
    x5              0.855      0.856 |     0.811      0.807 |     0.899      0.895
    x6              0.838      0.838 |     0.792      0.789 |     0.884      0.882
  speed =~                           |                       |
    x7              0.570      0.546 |     0.465      0.396 |     0.674      0.661
    x8              0.723      0.718 |     0.624      0.575 |     0.822      0.842
    x9              0.665      0.692 |     0.565      0.532 |     0.765      0.810
```

Output: Standardized Estimates of Covariances

-- Covariances:

	Estimate		Lower Bound (2.5%)		Upper Bound (97.5%)	
	Freq	Bayes	Freq	Bayes	Freq	Bayes
visual ~~						
textual	0.459	0.443	0.334	0.312	0.584	0.563
speed	0.471	0.459	0.328	0.297	0.613	0.634
textual ~~						
speed	0.283	0.275	0.148	0.138	0.418	0.405

Output: Standardized Estimates of Variances

-- Variances:

	Estimate		Lower Bound (2.5%)		Upper Bound (97.5%)	
	Freq	Bayes	Freq	Bayes	Freq	Bayes
.x1	0.404	0.421	0.238	0.234	0.571	0.600
.x2	0.821	0.813	0.722	0.708	0.920	0.919
.x3	0.662	0.650	0.537	0.521	0.788	0.796
.x4	0.275	0.280	0.200	0.209	0.350	0.366
.x5	0.269	0.268	0.194	0.198	0.344	0.348
.x6	0.298	0.297	0.221	0.222	0.374	0.378
.x7	0.676	0.702	0.557	0.567	0.794	0.848
.x8	0.477	0.484	0.334	0.291	0.620	0.670
.x9	0.558	0.522	0.425	0.344	0.691	0.717
visual	1.000	1.000	1.000	1.000	1.000	1.000
textual	1.000	1.000	1.000	1.000	1.000	1.000
speed	1.000	1.000	1.000	1.000	1.000	1.000

Output: Fit Statistics (Bayesian & Frequentist)

```
$freq
        chisq      df  pvalue    cfi  rmsea  rmsea.ci.lower  rmsea.ci.upper   srmr
       85.306  24.000   0.000  0.931  0.092           0.071           0.114  0.065


$bayes
        ppp         cfi    cfi.2.5%   cfi.97.5%       rmsea  rmsea.2.5% rmsea.97.5%
 0.00000000  0.09946339  0.91401706  0.94293958  0.09946339  0.09029012  0.11062469
```

Output: Fit Statistics (Information Theory)

```
$info
        chisq df pvalue       cfi     rmsea rmsea.ci.lower rmsea.ci.upper      srmr
10%  775.2136 24      0 0.2117886 0.3224726      0.3031798      0.3421804 0.1728616
20%  916.6404 24      0 0.2549585 0.3515195      0.3322339      0.3711878 0.1854028
30% 1031.0950 24      0 0.2887574 0.3733760      0.3540939      0.3930196 0.1951621
40% 1144.1069 24      0 0.3194740 0.3937684      0.3744885      0.4133920 0.2038494
50% 1250.9533 24      0 0.3501582 0.4121214      0.3928428      0.4317290 0.2122930
60% 1369.4100 24      0 0.3824575 0.4315573      0.4122796      0.4511498 0.2214058
70% 1503.3272 24      0 0.4193888 0.4525257      0.4332484      0.4721038 0.2313275
80% 1675.1744 24      0 0.4651925 0.4780877      0.4588104      0.4976502 0.2440951
90% 1971.3581 24      0 0.5298230 0.5191992      0.4999212      0.5387405 0.2653284
```

cosme Package in R

This package is still under development

- An ongoing large-scale simulation will provide guidelines for applied FP analysis
  - e.g., recommendations about # of random data sets, optimal estimation/optimization settings
  - These guidelines will inform the default settings in cosme package
- Plotting functions to be added
- Available later this year!

Please complete our survey (link in OSF wiki):

https://osf.io/rquts/



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Thank you!

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